



US012416408B2

(12) **United States Patent**
Benjamin et al.

(10) **Patent No.:** **US 12,416,408 B2**

(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **COMBUSTOR SIZE RATING FOR A GAS TURBINE ENGINE USING HYDROGEN FUEL**

(71) Applicant: **General Electric Company**,
Schenectady, NY (US)

(72) Inventors: **Michael A. Benjamin**, Cincinnati, OH
(US); **Manampathy G. Giridharan**,
Mason, OH (US)

(73) Assignee: **General Electric Company**, Evendale,
OH (US)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/673,532**

(22) Filed: **May 24, 2024**

(65) **Prior Publication Data**

US 2024/0310048 A1 Sep. 19, 2024

Related U.S. Application Data

(63) Continuation-in-part of application No. 17/457,559,
filed on Dec. 3, 2021, now Pat. No. 12,078,100.

(51) **Int. Cl.**
F02C 3/22 (2006.01)
F23R 3/14 (2006.01)

(52) **U.S. Cl.**
CPC . **F23R 3/14** (2013.01); **F02C 3/22** (2013.01)

(58) **Field of Classification Search**
CPC F02C 3/20; F02C 3/22; F05D 2250/80;
F23R 3/283; F23R 3/14

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,375,150 A	3/1983	Nikiforakis
4,794,754 A	1/1989	Shekleton et al.
5,109,670 A	5/1992	Harshman
5,307,633 A	5/1994	Koemer et al.

(Continued)

FOREIGN PATENT DOCUMENTS

CN	111553044 A	8/2020
CN	112231903 A	1/2021
CN	212537825 U	2/2021

OTHER PUBLICATIONS

Lefebvre et al., "Gas Turbine Combustion: Alternative Fuels and
Emissions," CRC Press, 3rd Edition (Apr. 26, 2010).

(Continued)

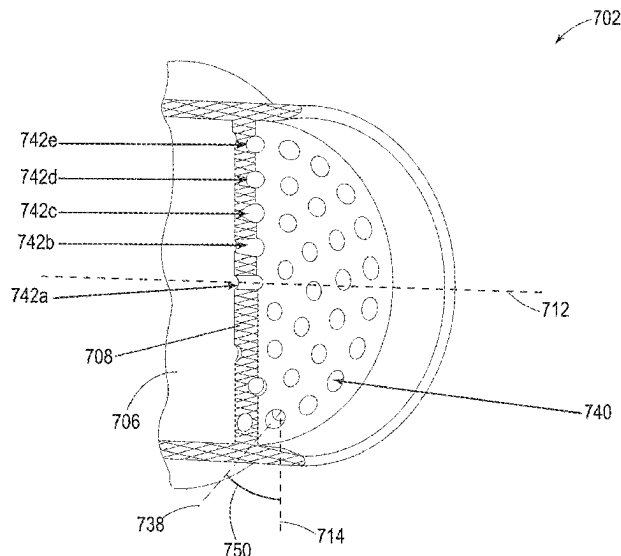
Primary Examiner — Gerald L Sung

(74) *Attorney, Agent, or Firm* — McGarry Bair PC

(57) **ABSTRACT**

A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length and a burner dome height. The combustion chamber is configured to combust a mixture of the hydrogen fuel flow and the compressed air flow. The combustion chamber can be characterized by a combustor size rating between one inch and seven inches. In more detail, the combustion chamber can be characterized by the combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, in which the combustor size rating is a function of the core air flow parameter.

15 Claims, 23 Drawing Sheets



(56)

References Cited

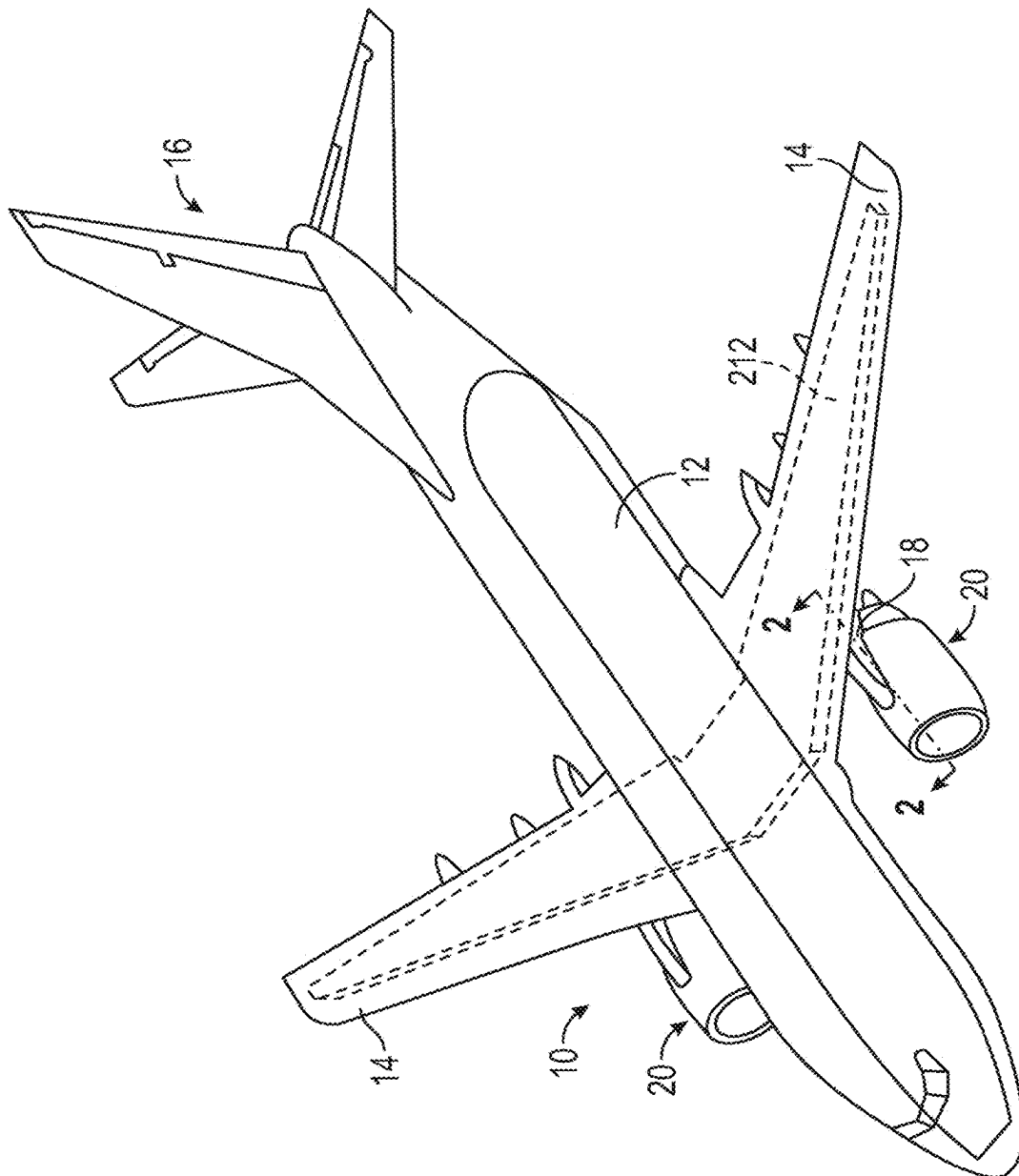
U.S. PATENT DOCUMENTS

5,335,502	A	8/1994	Roberts, Jr. et al.
6,810,673	B2	11/2004	Snyder
7,014,835	B2	3/2006	Mathias et al.
7,900,457	B2	3/2011	Patterson et al.
8,726,626	B2	5/2014	Spooner
9,581,085	B2	2/2017	Bartz et al.
10,317,080	B2	6/2019	Tu, Jr. et al.
10,775,047	B2	9/2020	Horikawa et al.
11,674,443	B2	6/2023	McCurdy Gibson et al.
11,732,652	B2	8/2023	Sibbach et al.
2005/0287407	A1	12/2005	Bushko
2008/0078182	A1	4/2008	Evulet
2008/0256924	A1	10/2008	Pederson et al.
2009/0199563	A1	8/2009	Chen
2010/0293959	A1	11/2010	Remy et al.
2014/0090392	A1	4/2014	Meisner et al.
2015/0323189	A1	11/2015	Jeney et al.
2017/0298817	A1	10/2017	Horiuchi et al.
2019/0017441	A1	1/2019	Venkatesan et al.
2020/0095956	A1	3/2020	Ortelt et al.
2021/0071590	A1	3/2021	Beita et al.
2021/0071870	A1	3/2021	Bulat
2022/0030631	A1	1/2022	Sibbach et al.
2022/0307428	A1	9/2022	Sibbach et al.

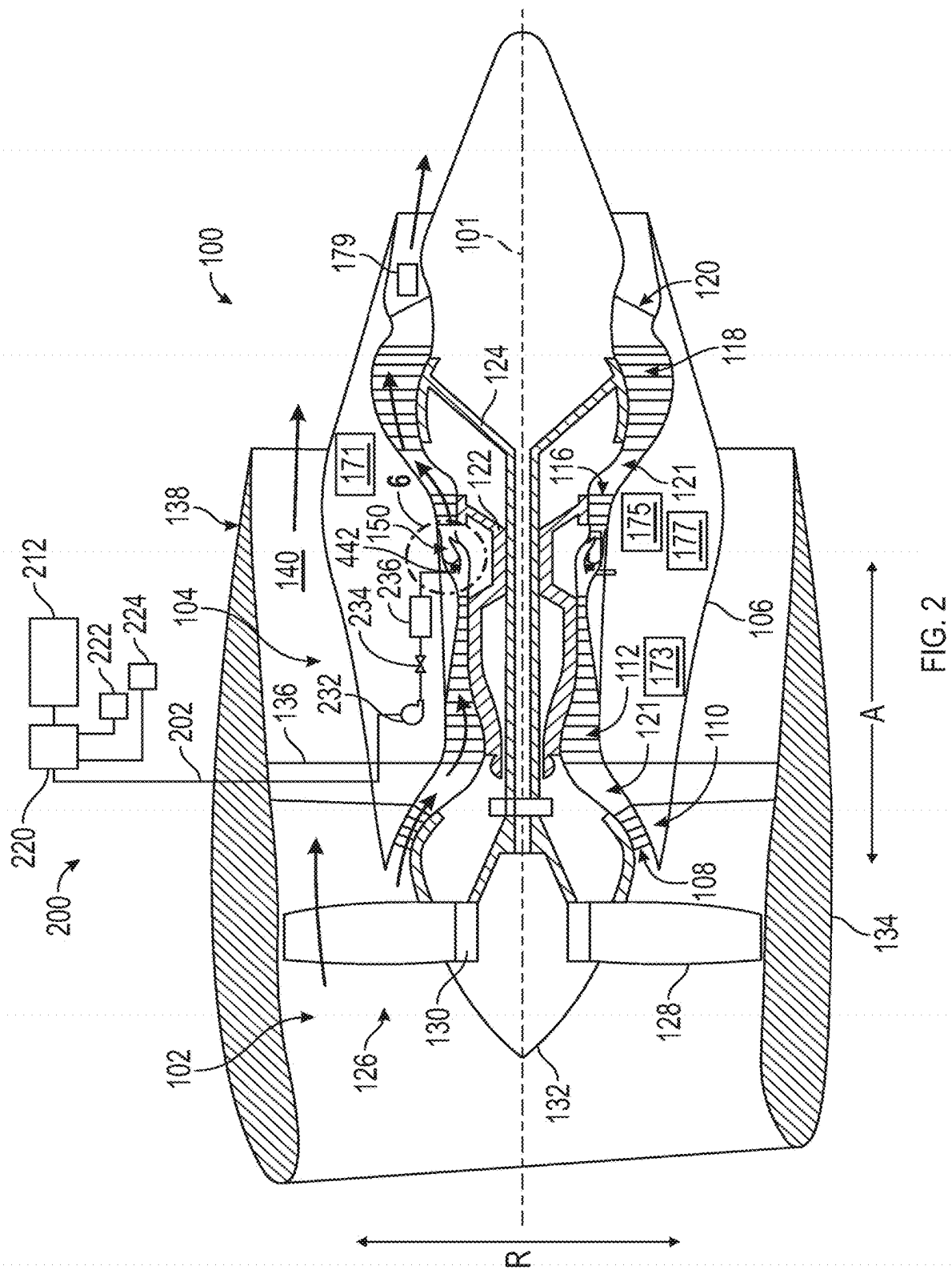
OTHER PUBLICATIONS

Morgan et al., "Longitudinal Instability Limits With a Variable-Length Hydrogen-Oxygen Combustor," Nasa Technical Note, Lewis Research Center, National Aeronautics and Space Administration (1971).

Hua et al., "Numerical simulation of combustion of hydrogen air mixture in micro-scaled chamber. Part I: Fundamental Study" Mar. 31, 2005 (Year: 2005).



५७



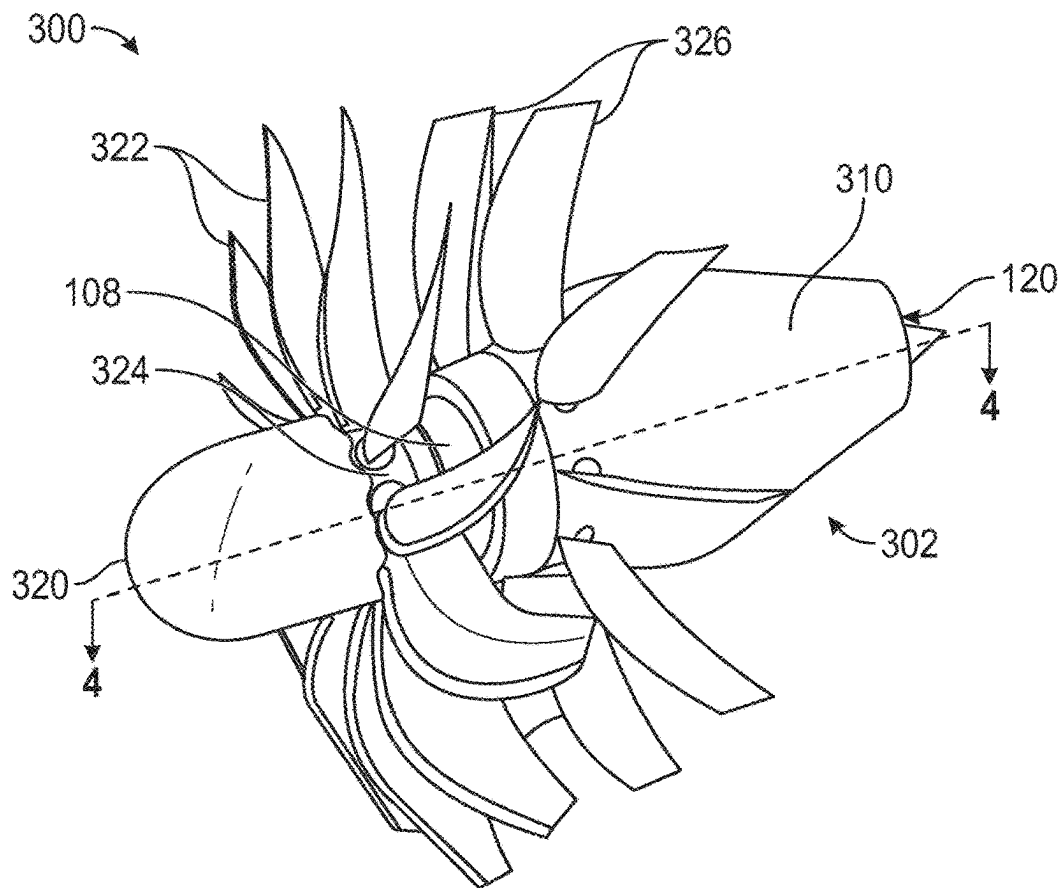


FIG. 3

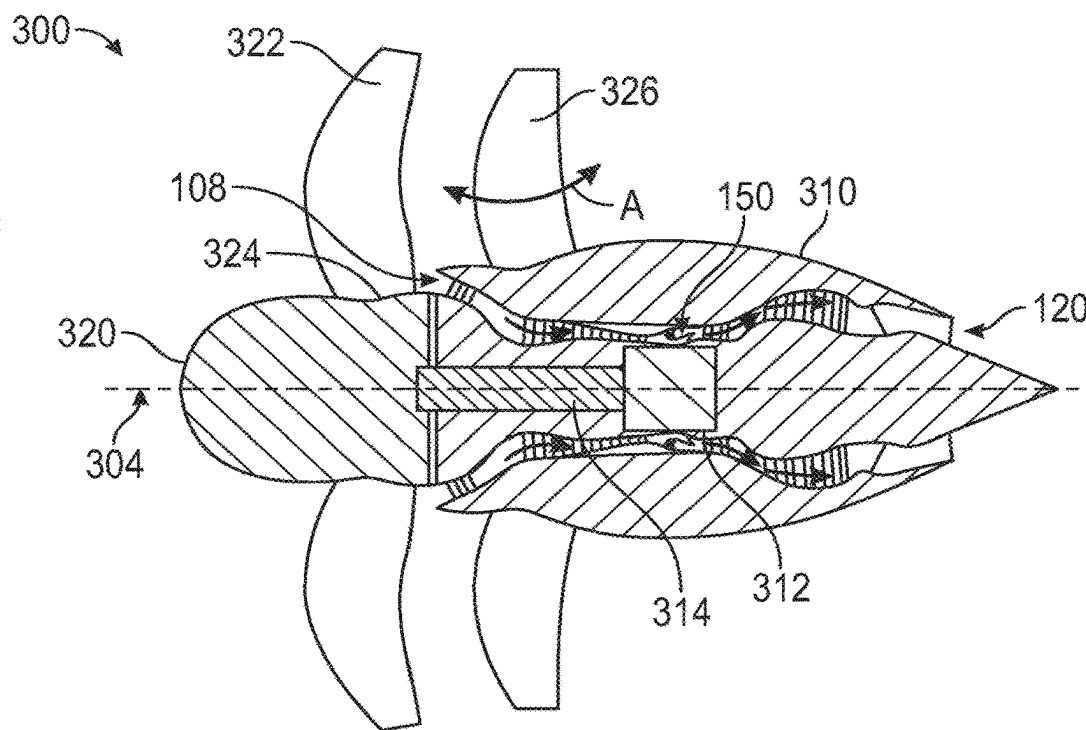


FIG. 4

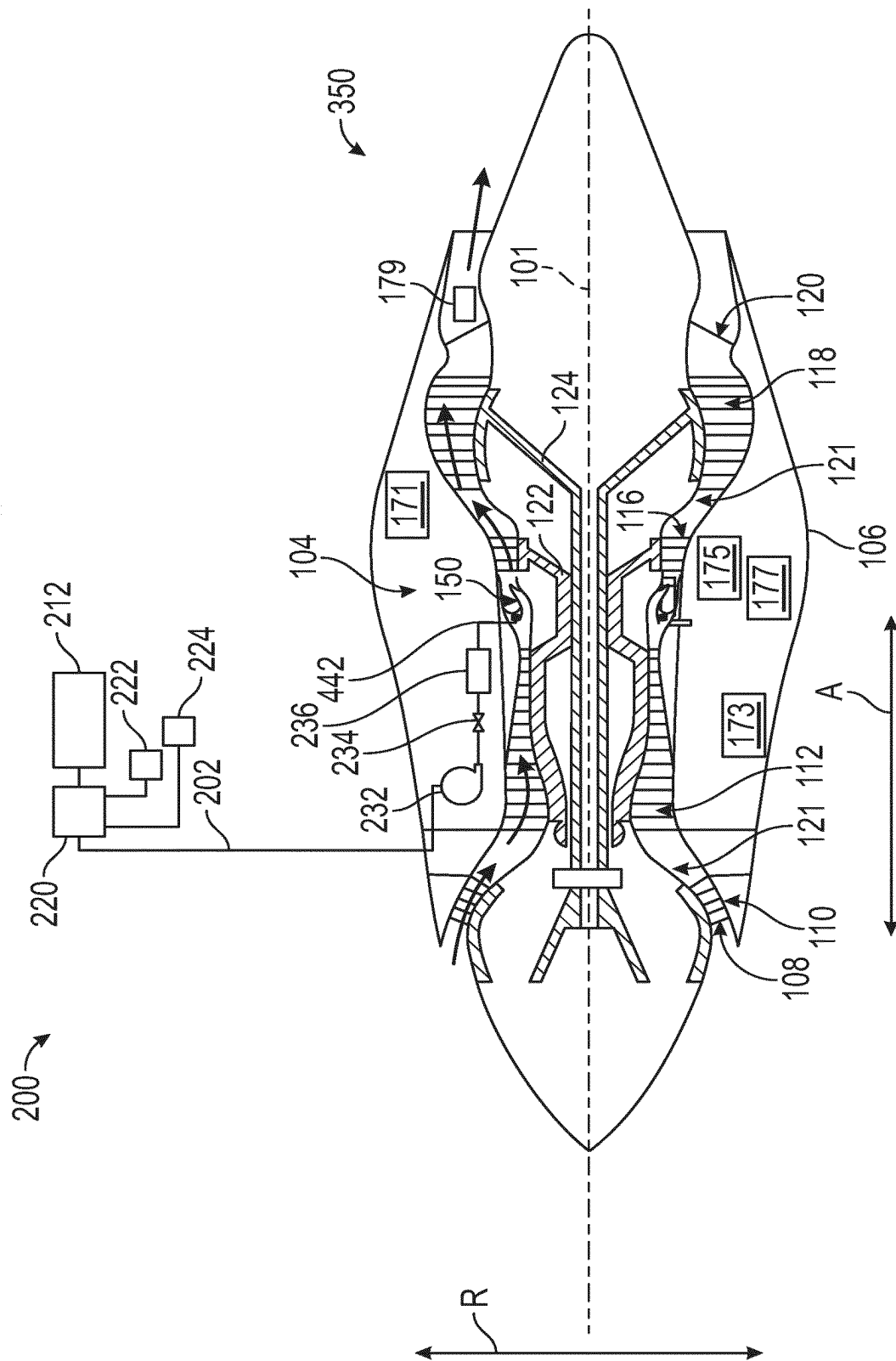


FIG. 5

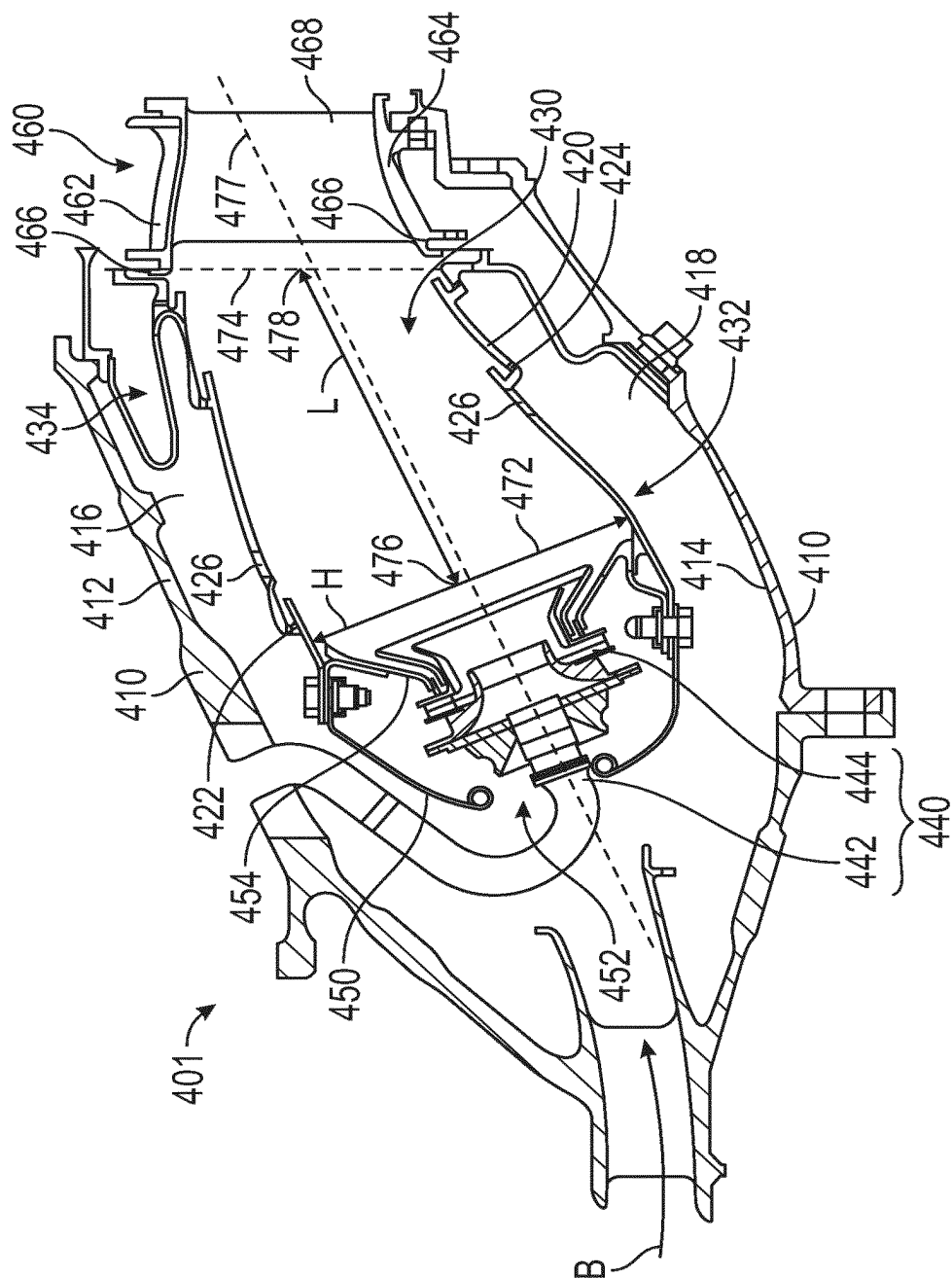
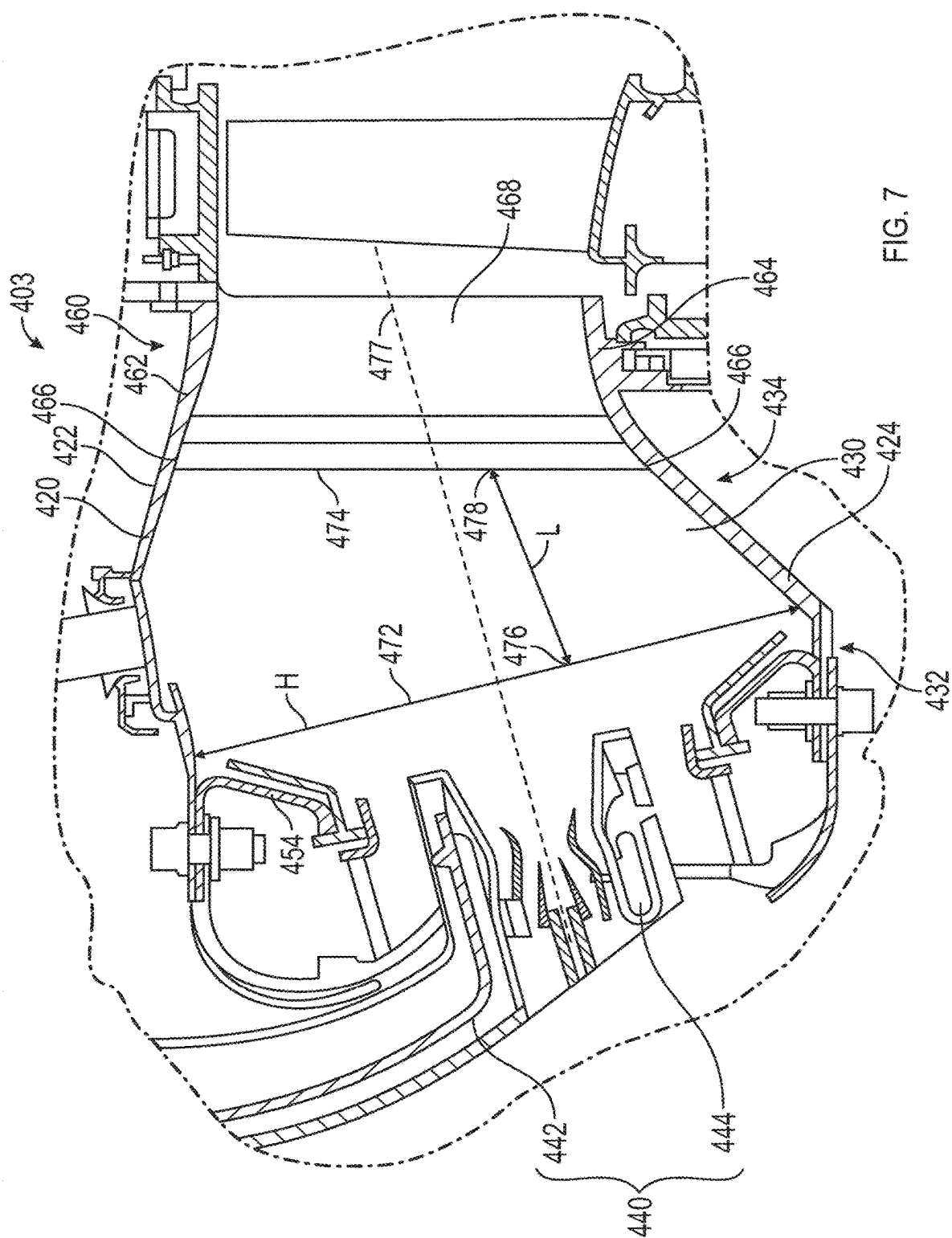
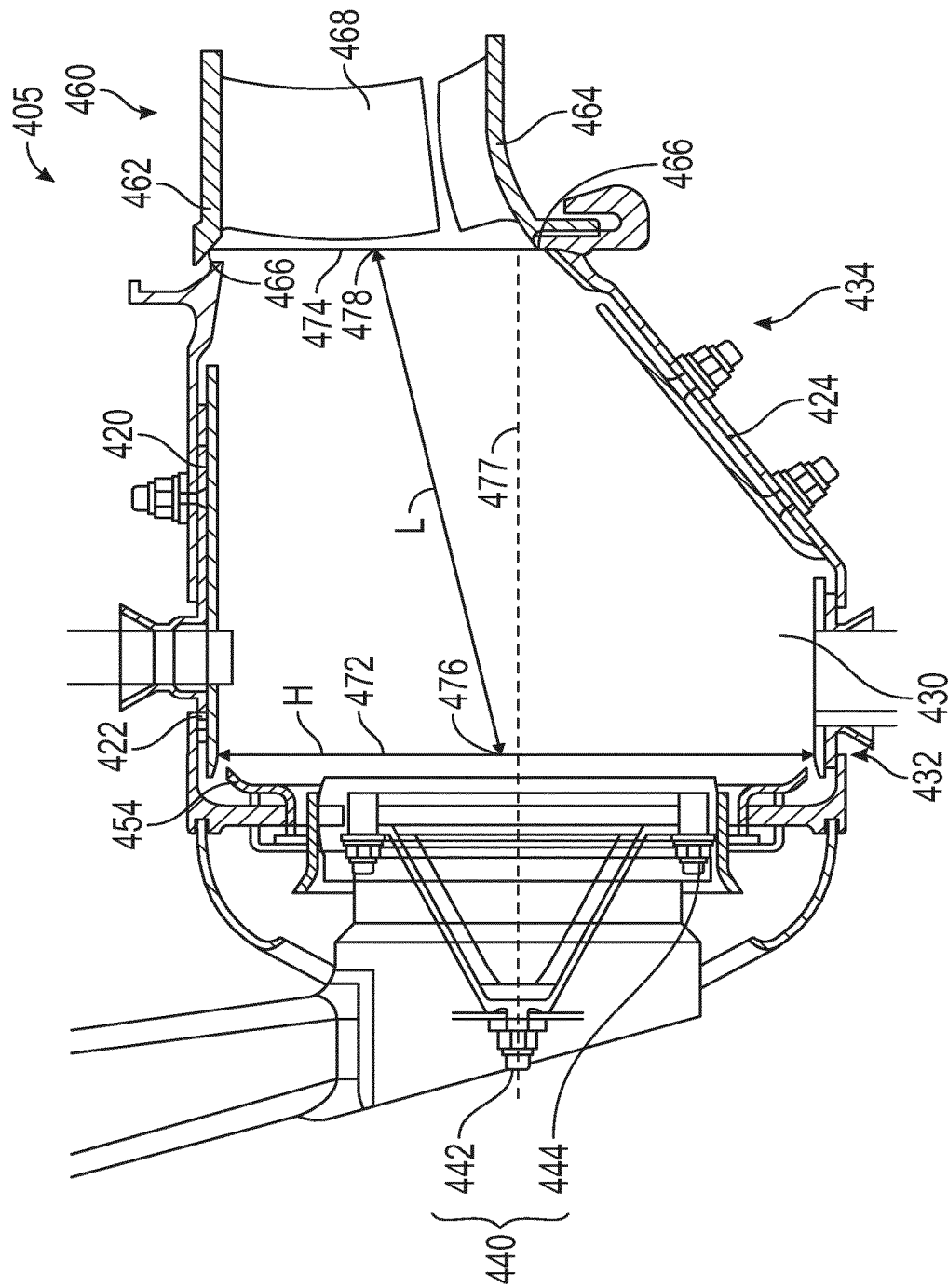


FIG. 6




$$\frac{\infty}{F/G}$$

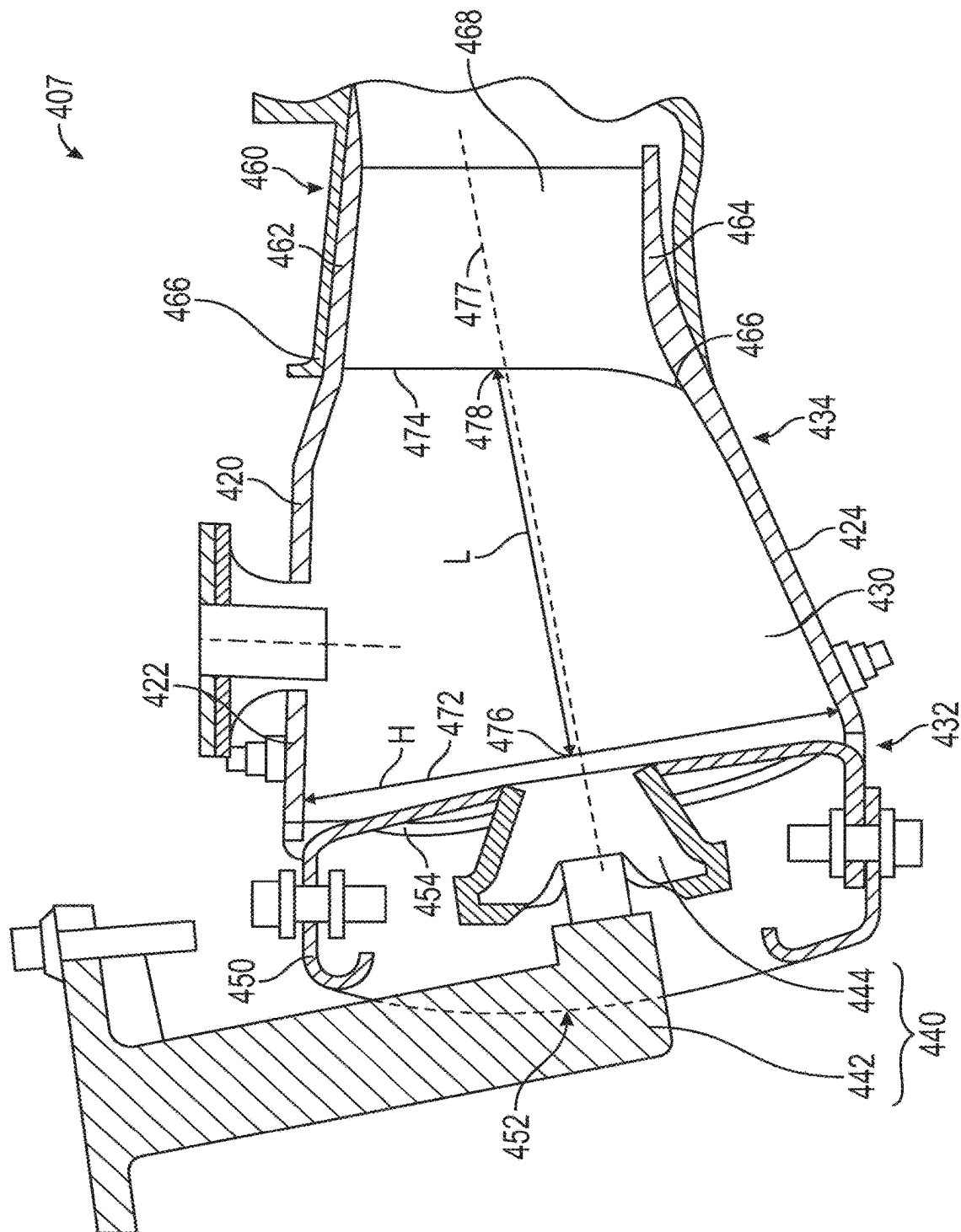


FIG. 9

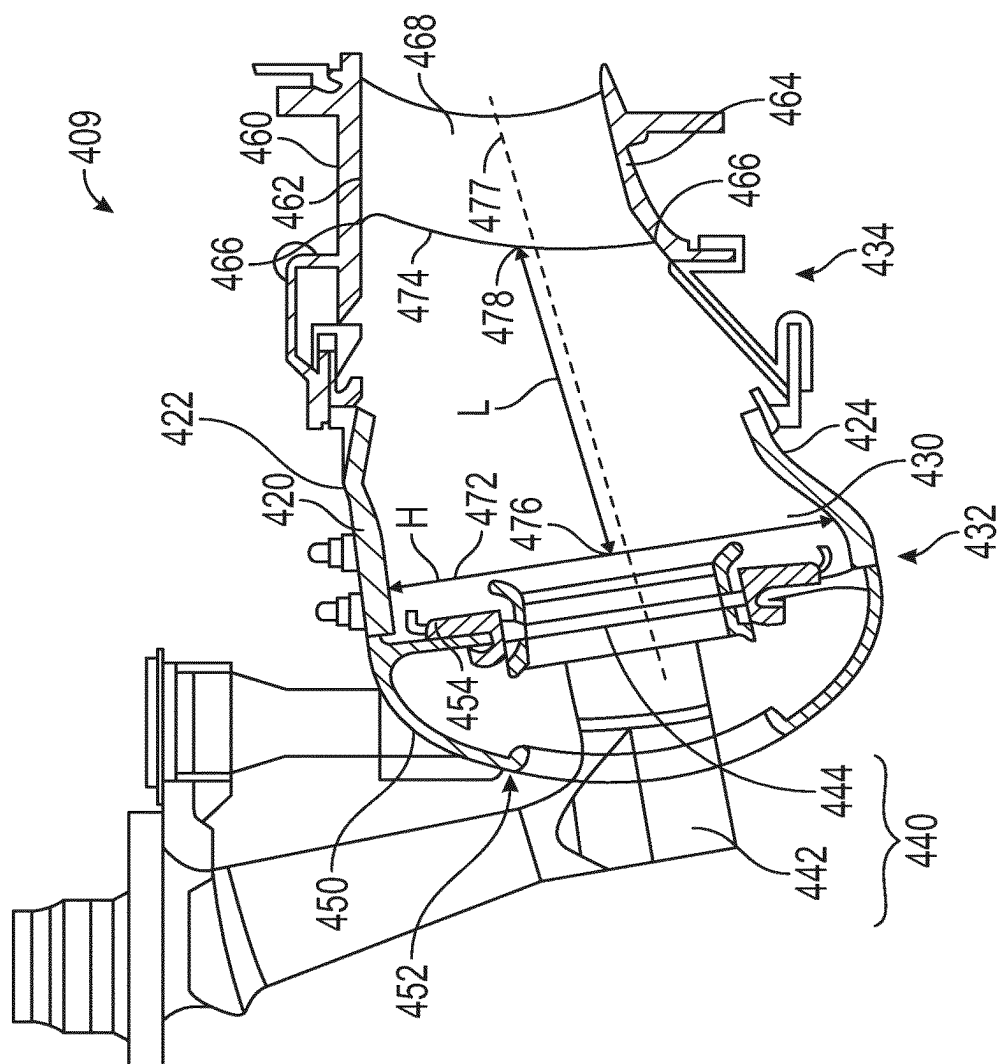


FIG. 10

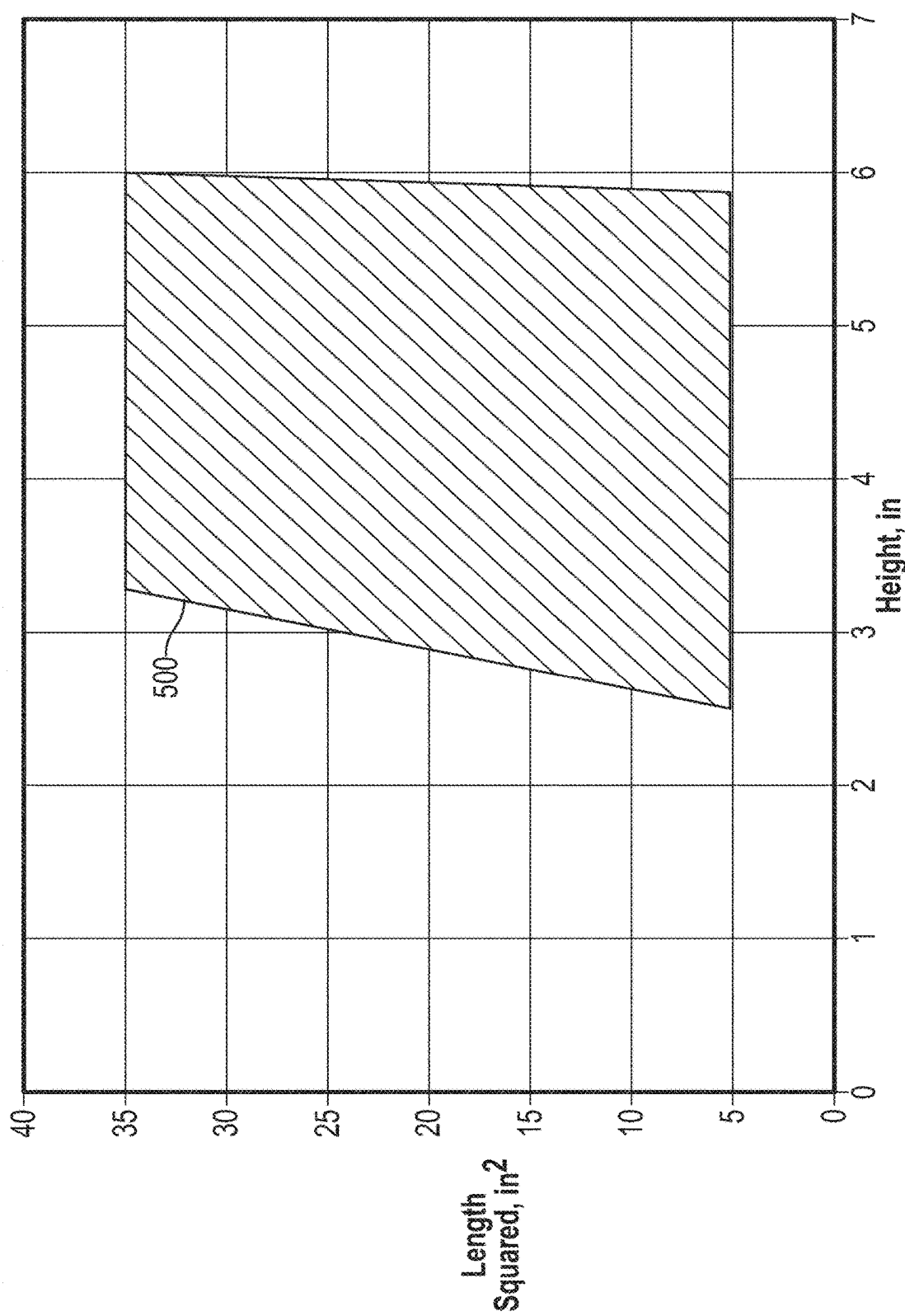


FIG. 11

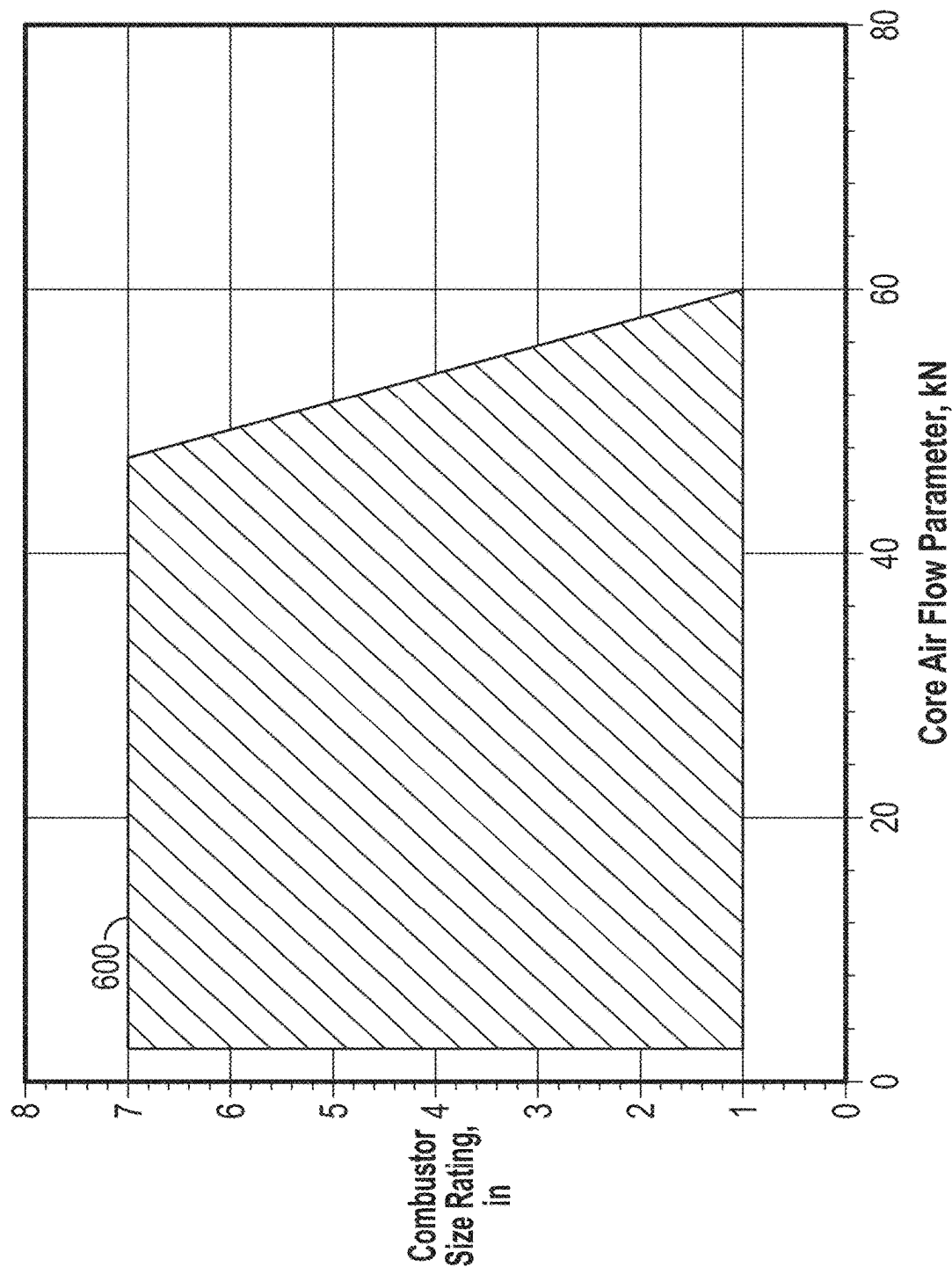


FIG. 12

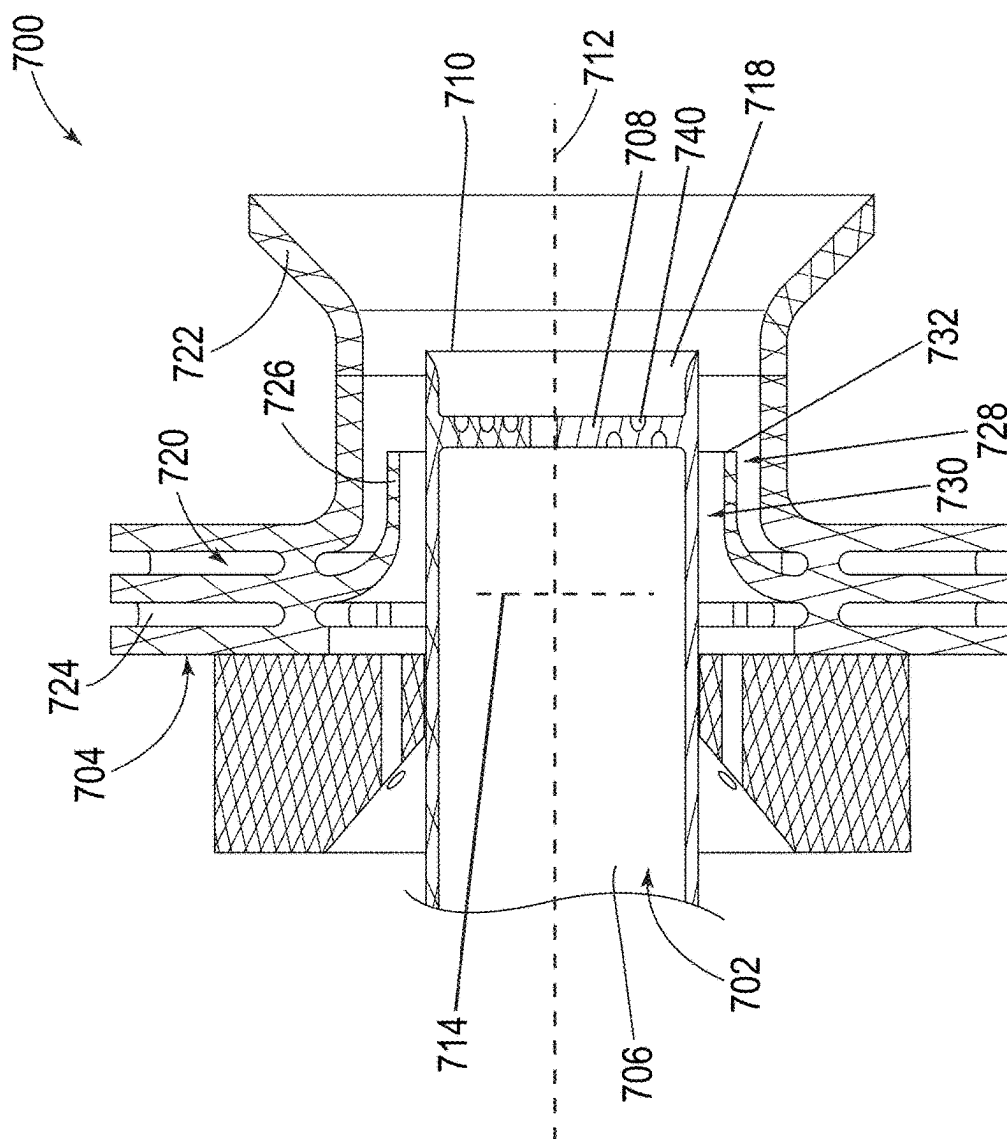


FIG. 13

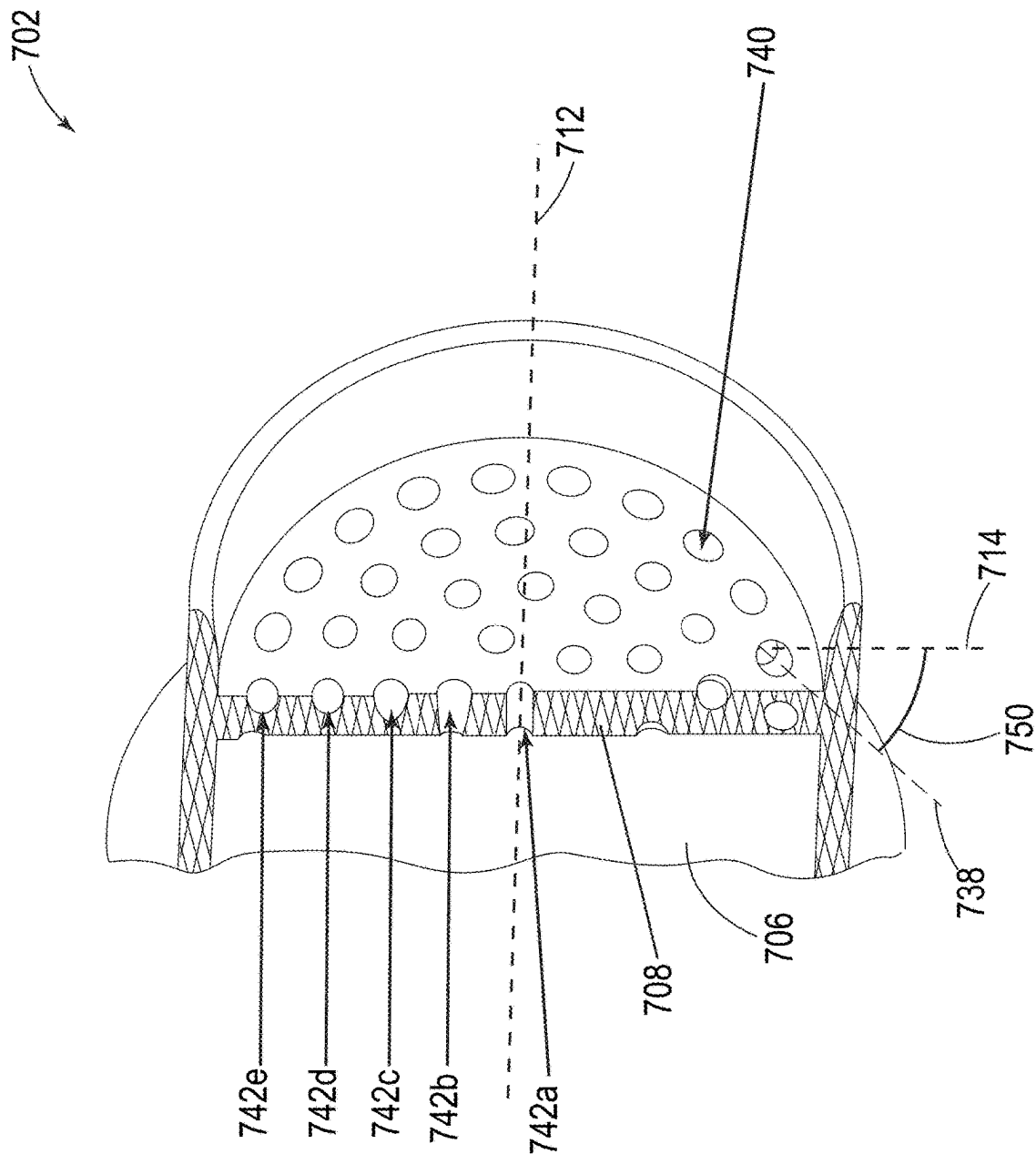


FIG. 14

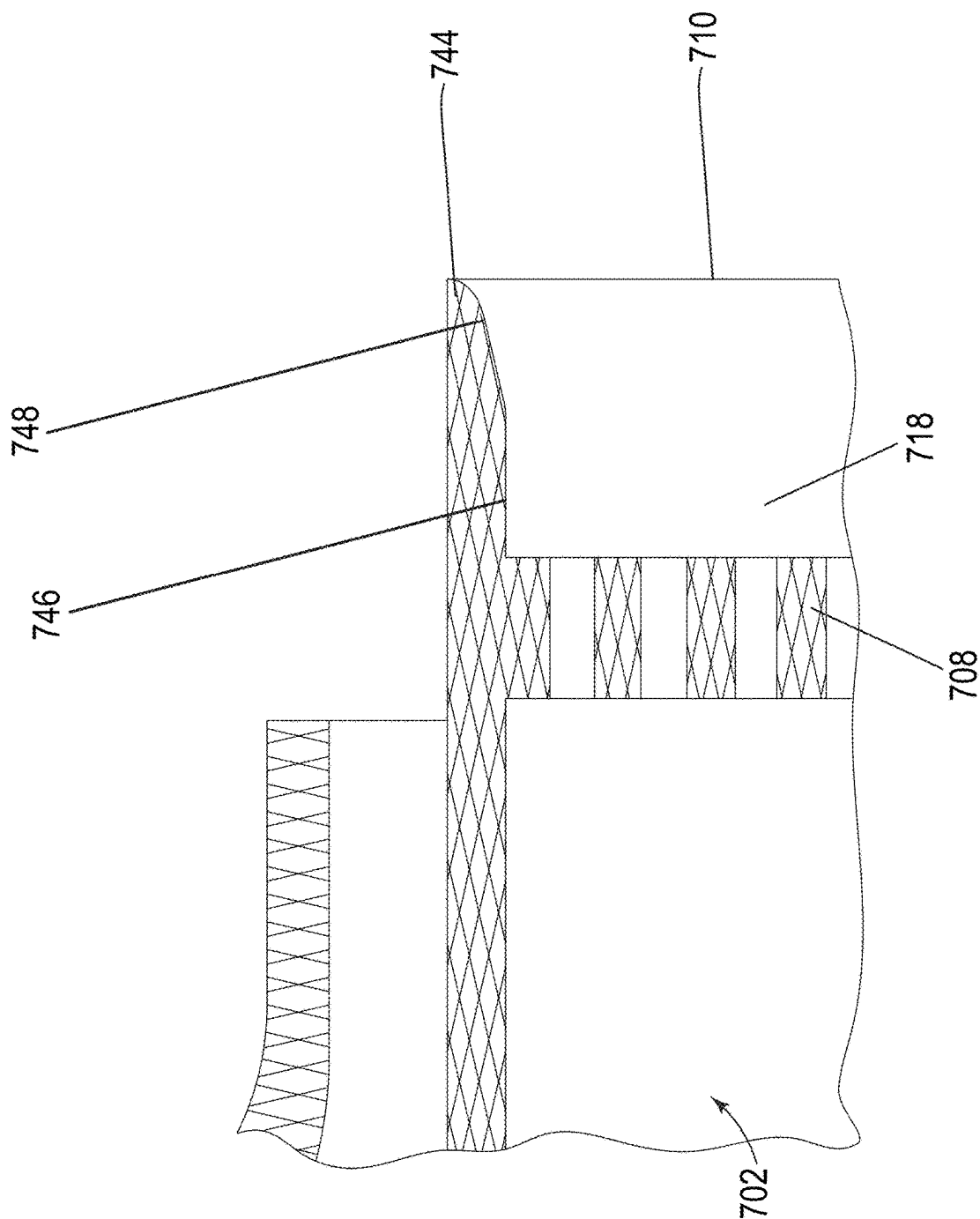


FIG. 15

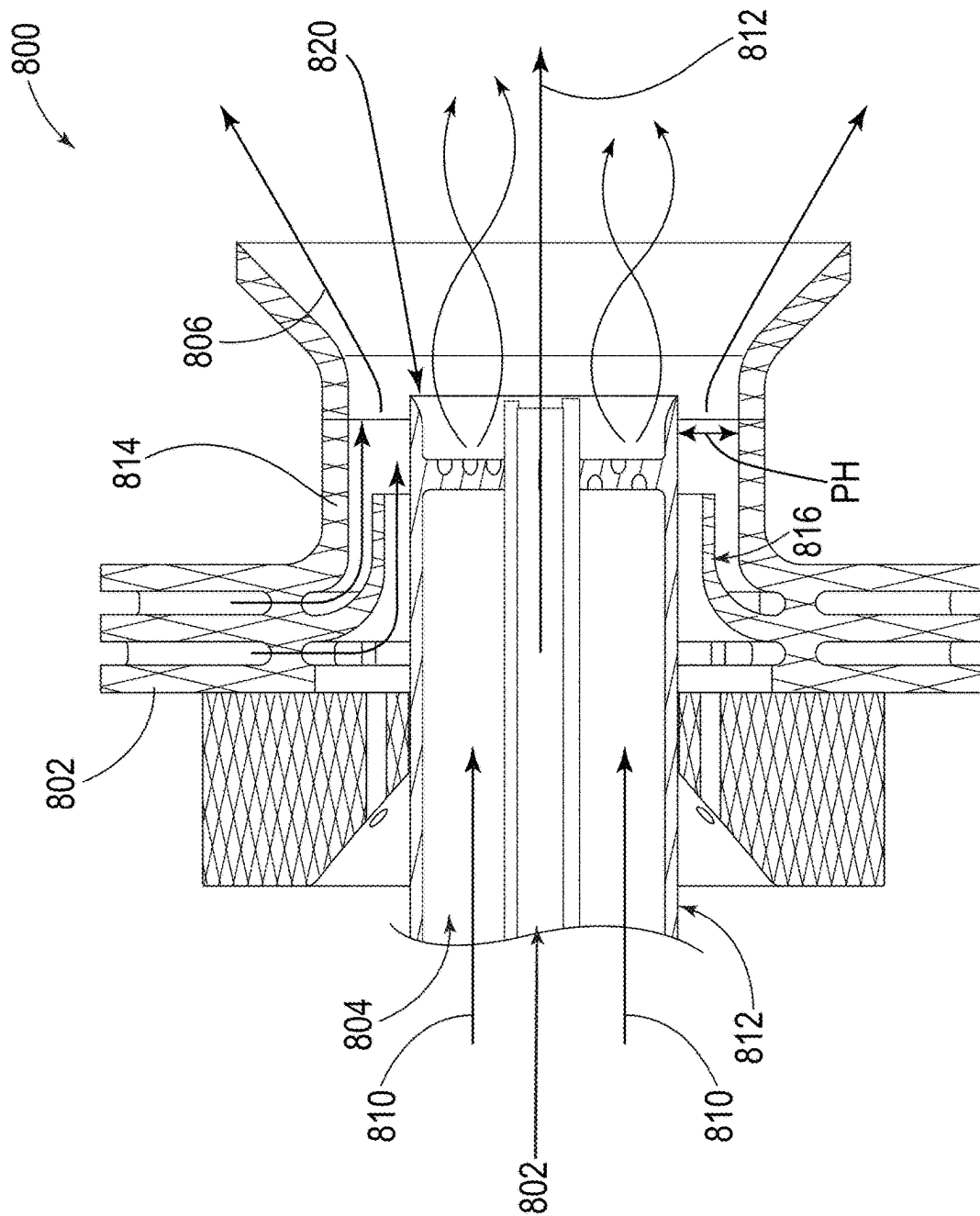


FIG. 16

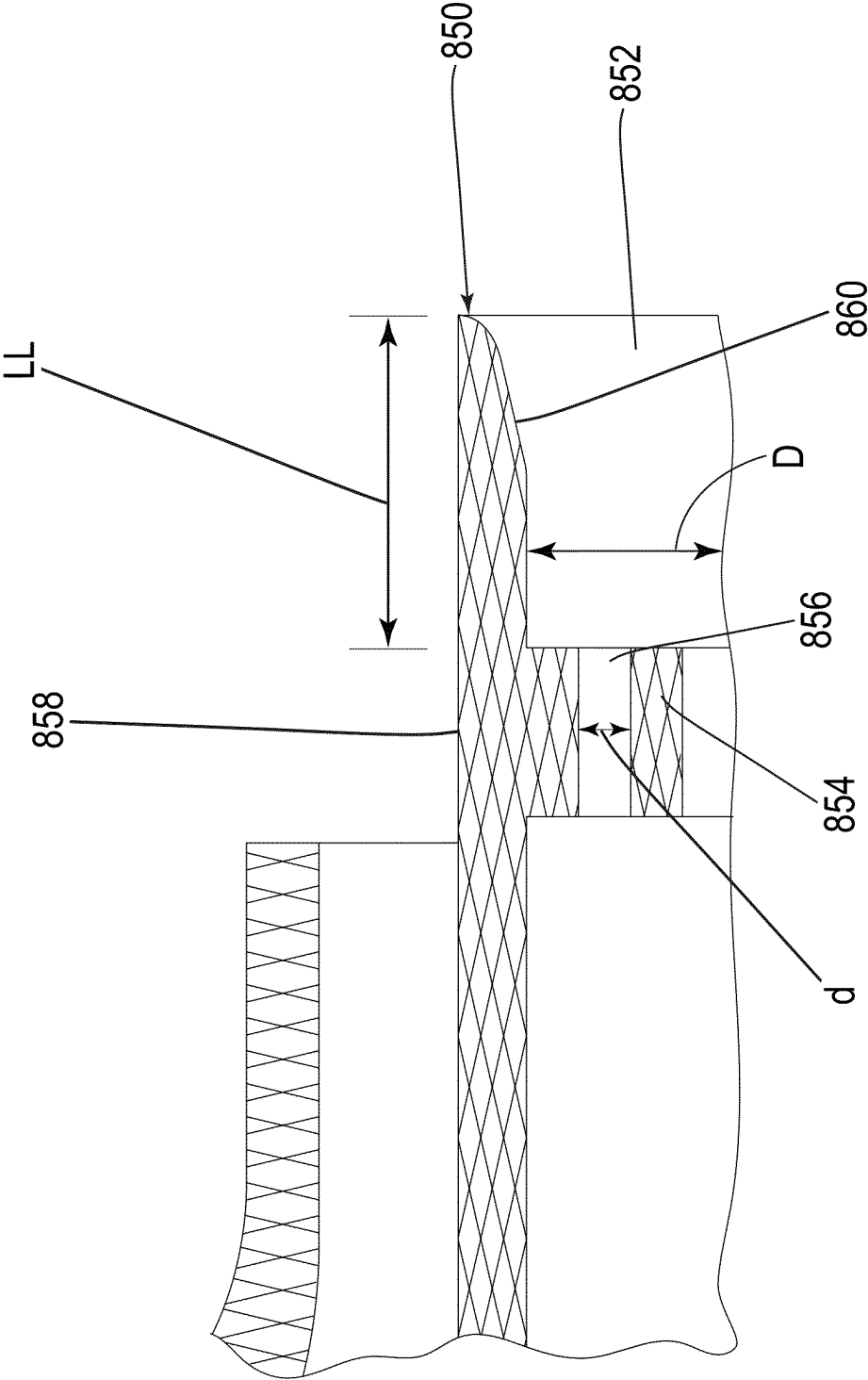


FIG. 17

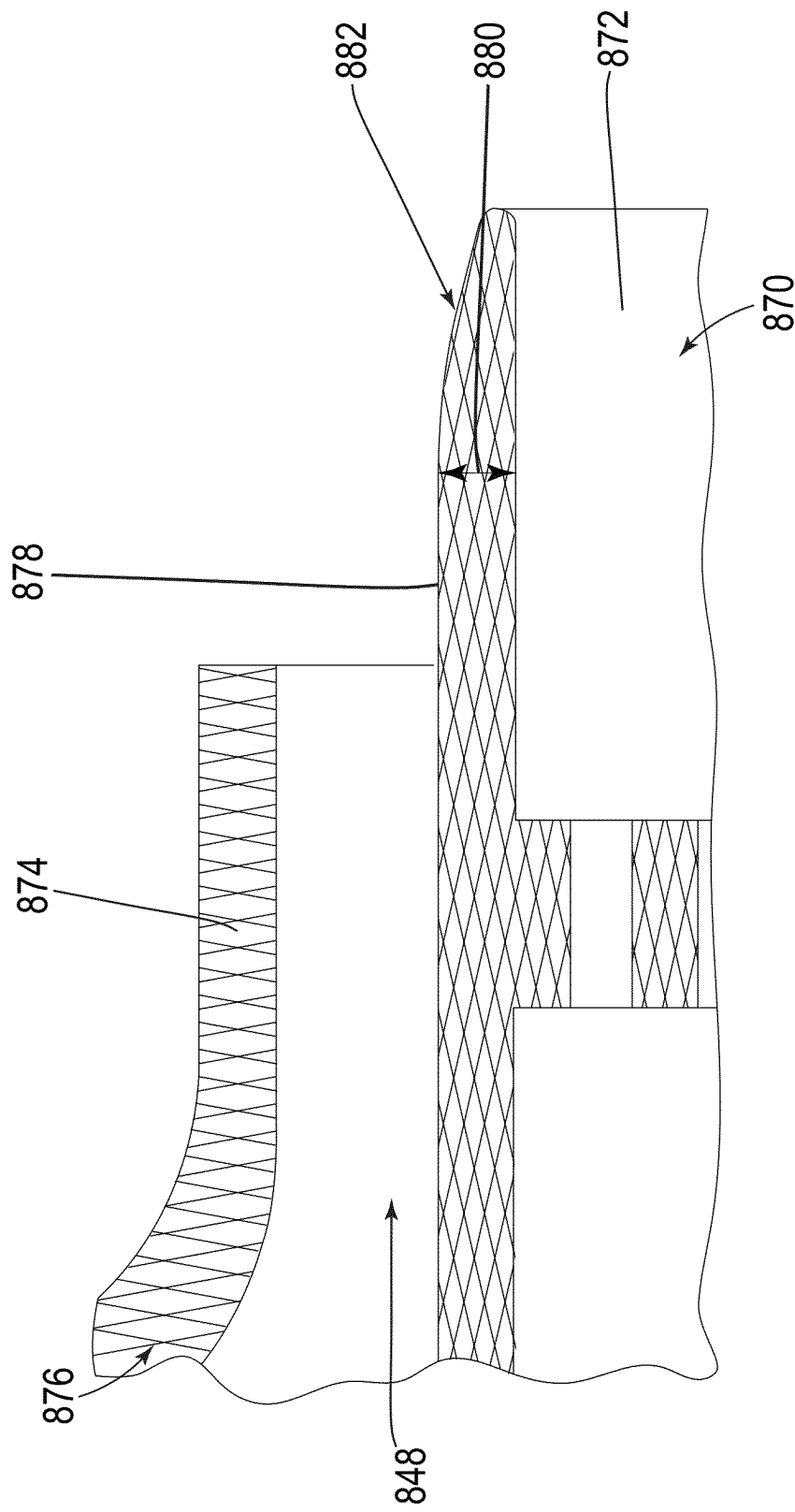


FIG. 18

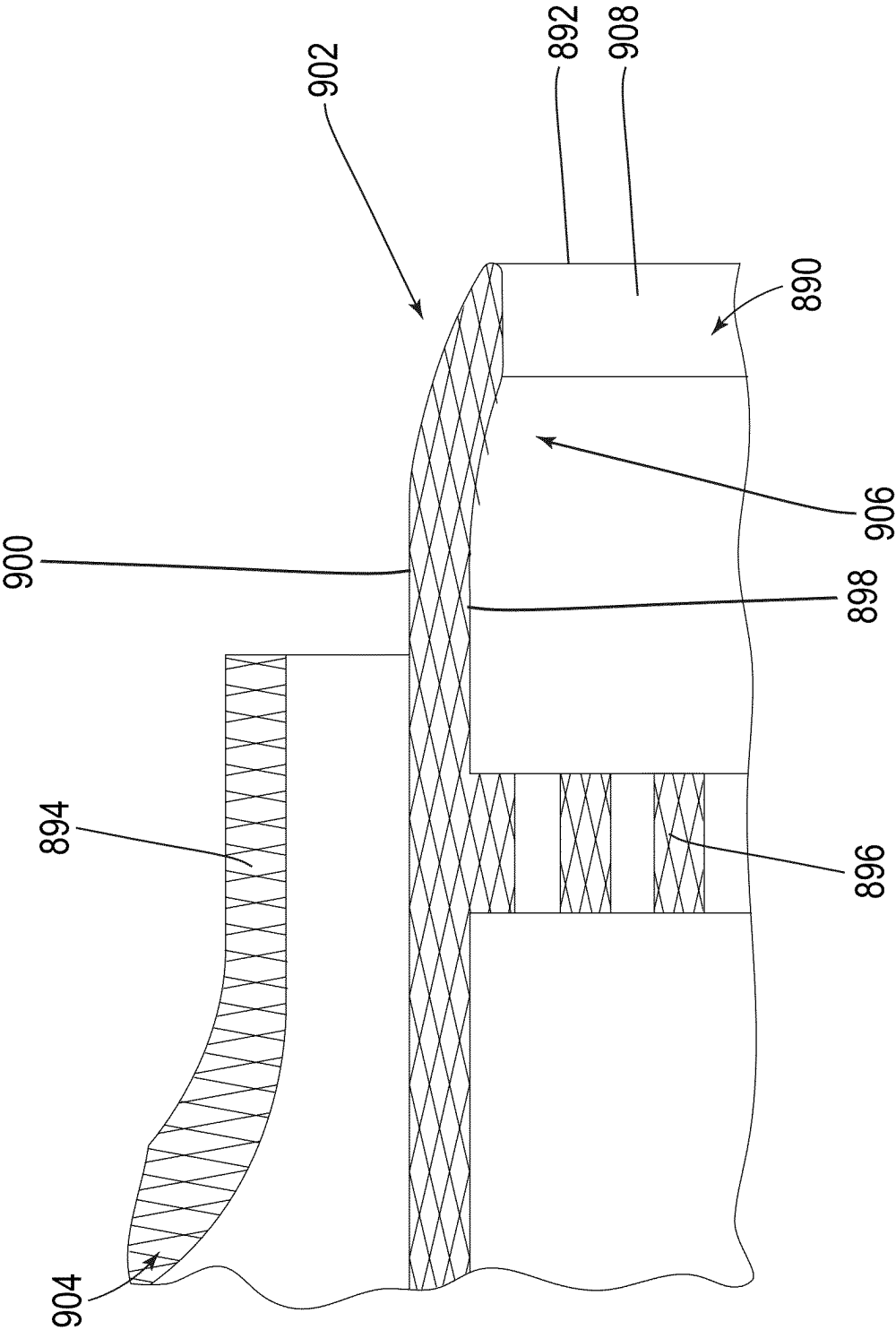


FIG. 19

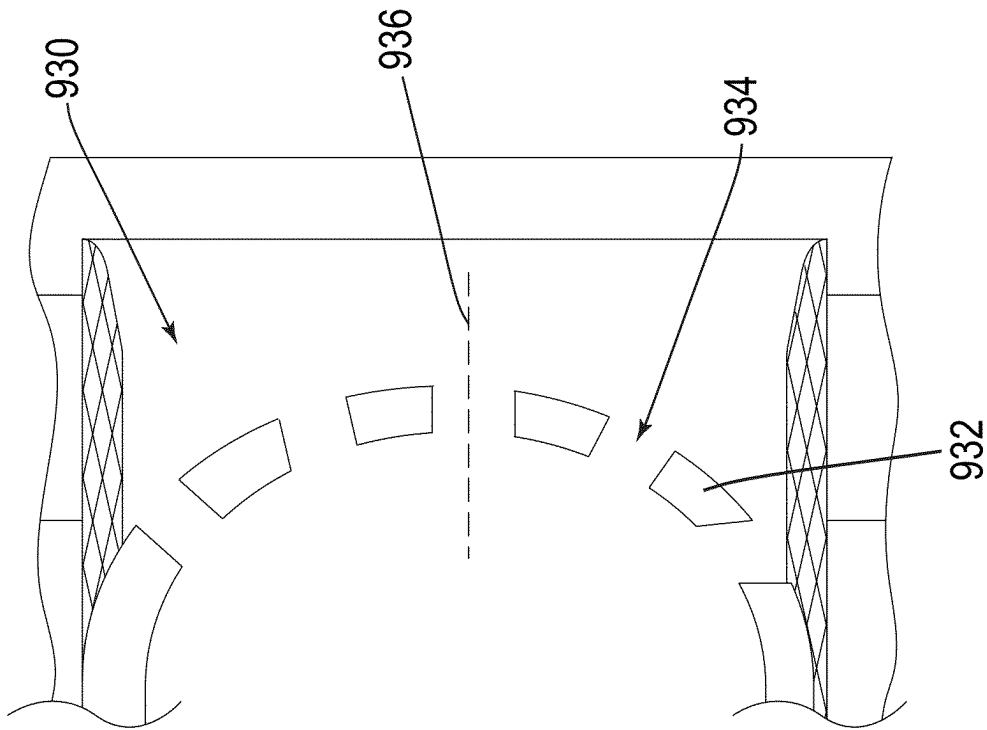


FIG. 20

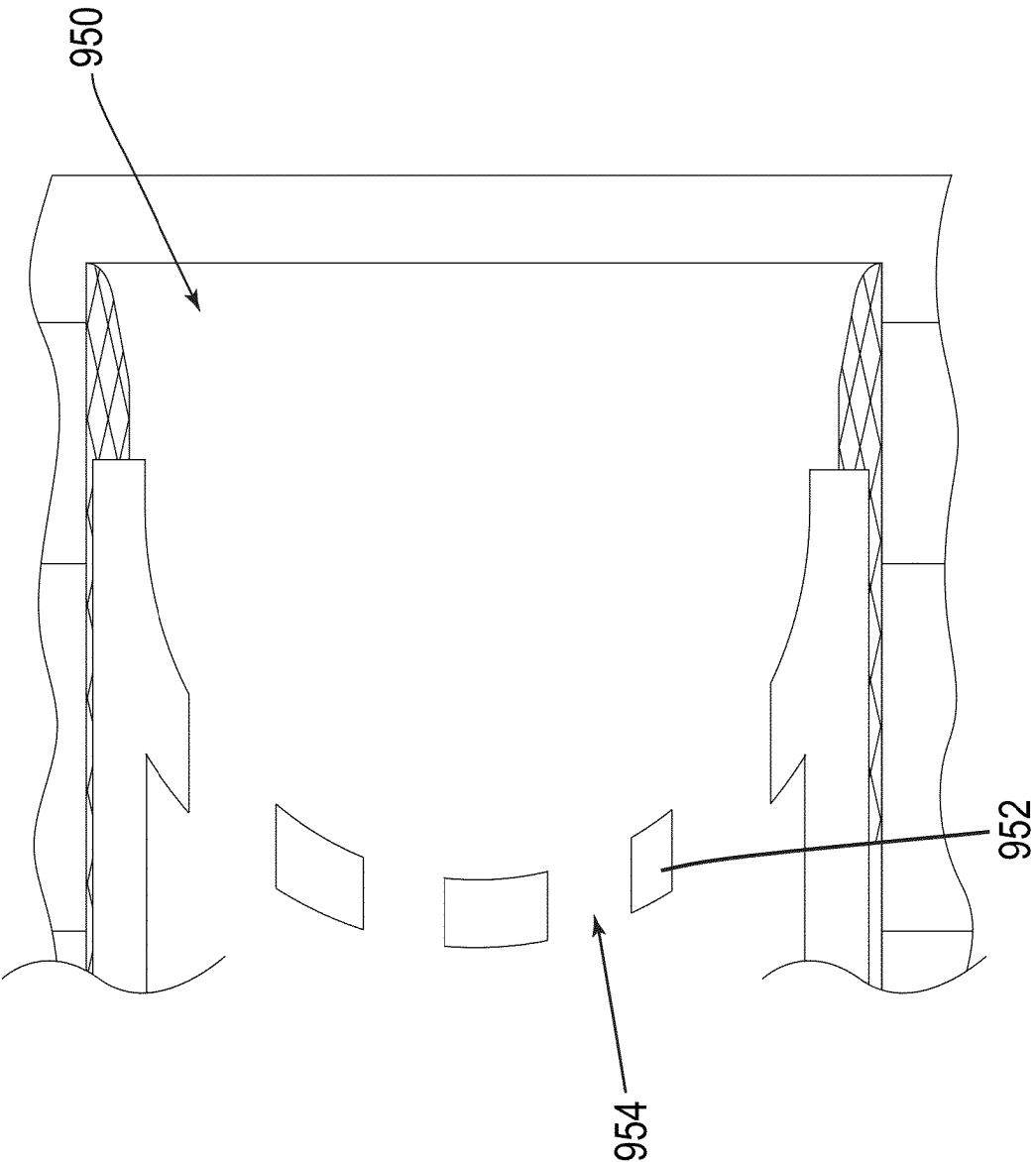


FIG. 21

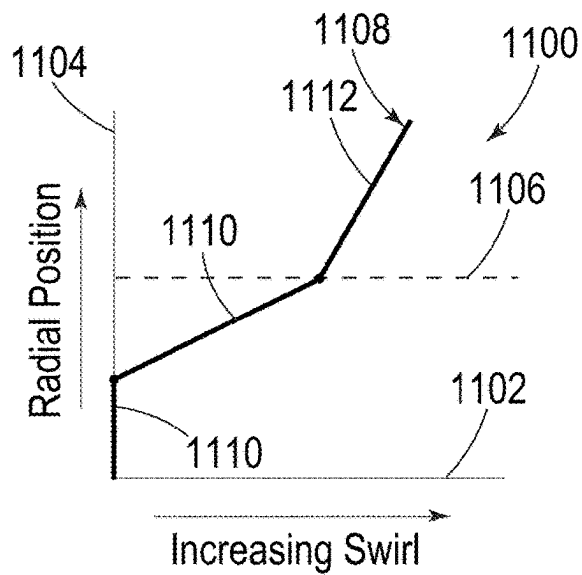


FIG. 22

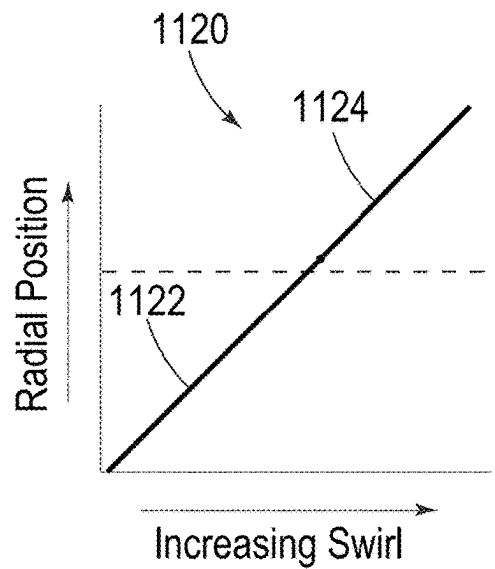


FIG. 23

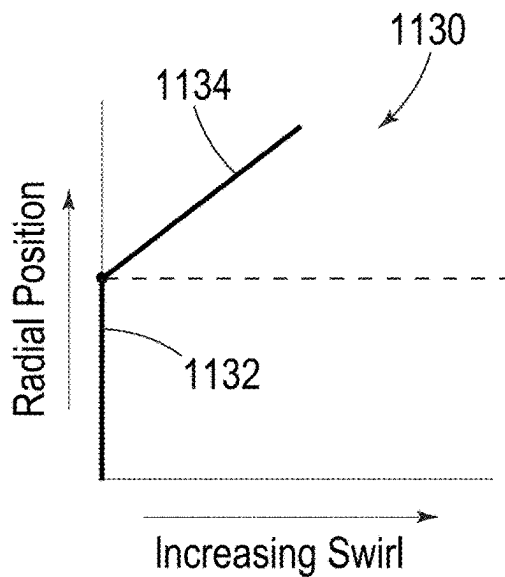


FIG. 24

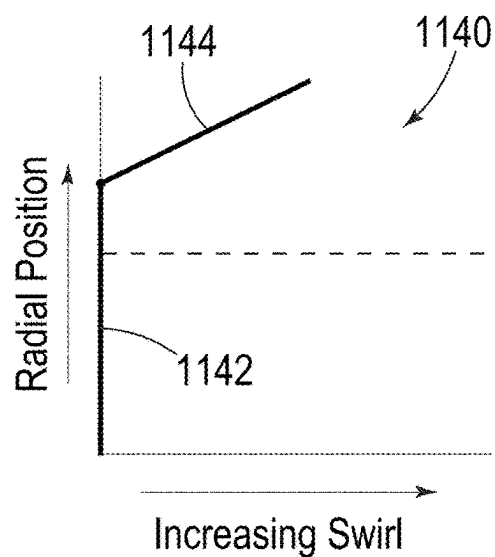


FIG. 25

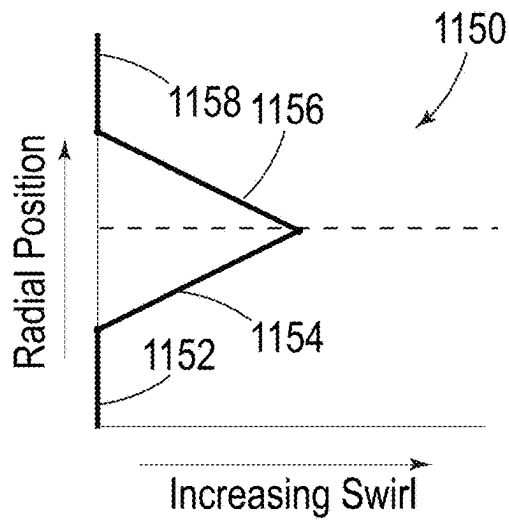


FIG. 26

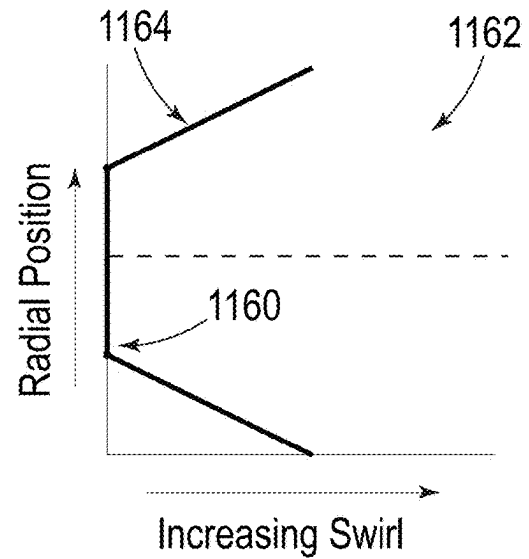


FIG. 27

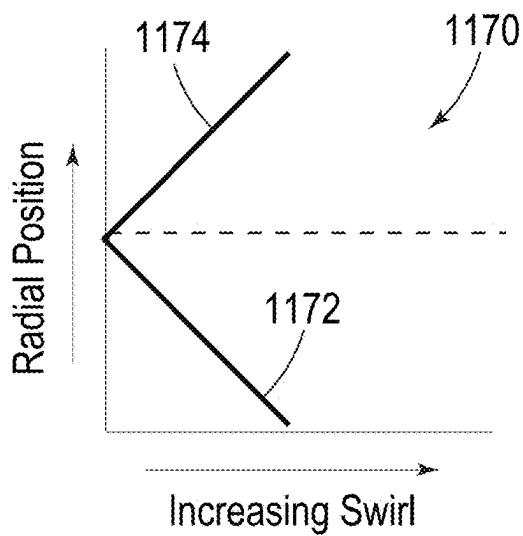


FIG. 28

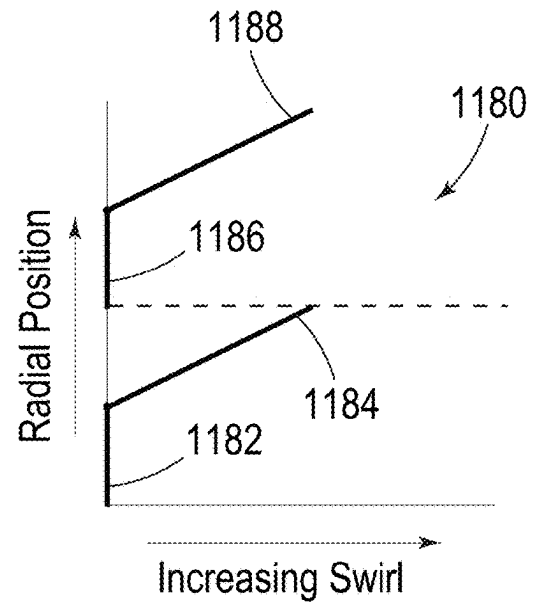


FIG. 29

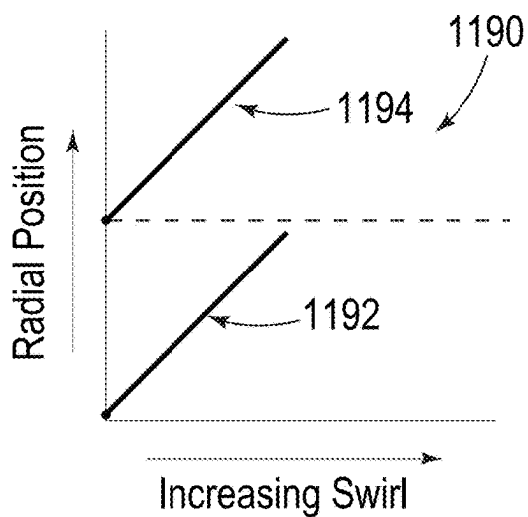


FIG. 30

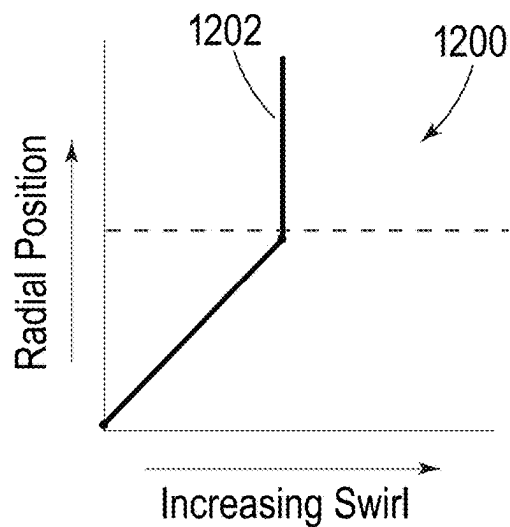


FIG. 31

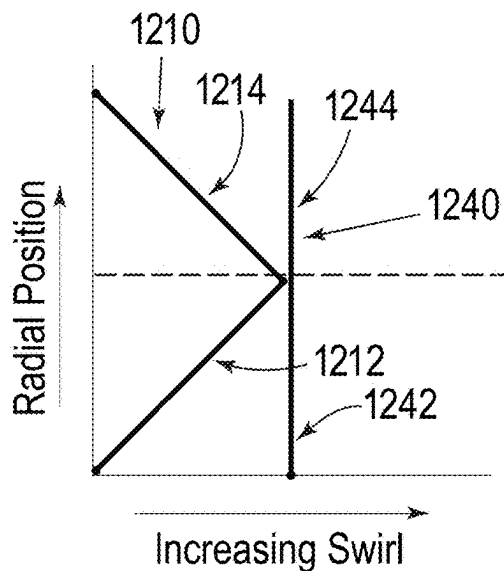


FIG. 32

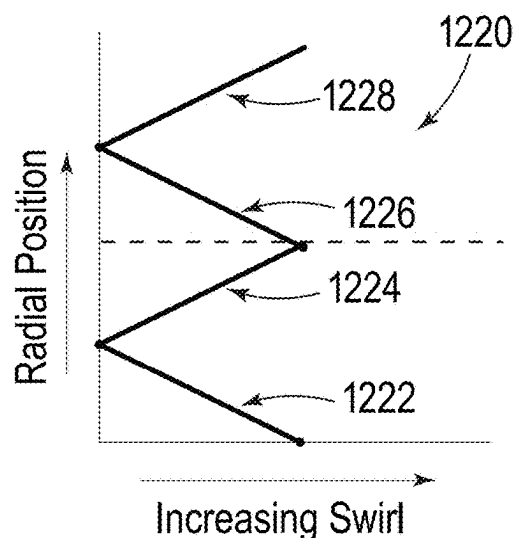


FIG. 33

1

COMBUSTOR SIZE RATING FOR A GAS TURBINE ENGINE USING HYDROGEN FUEL

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a continuation-in-part of U.S. application Ser. No. 17/457,559, filed Dec. 3, 2021, now allowed, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present disclosure relates to a combustor for a gas turbine engine using hydrogen fuel, and in particular, for a gas turbine engine for aircraft.

BACKGROUND

The propulsion system for commercial aircraft typically includes one or more aircraft engines, such as turbofan jet engines. The aircraft engine(s) may be mounted to a respective one of the wings of the aircraft, such as in a suspended position beneath the wing using a pylon. These engines may be powered by aviation turbine fuel, which is typically a combustible hydrocarbon liquid fuel, such as a kerosene-type fuel, having a desired carbon number and carbon to hydrogen ratio. Such fuel produces carbon dioxide emissions upon combustion and improvements to reduce such carbon dioxide emissions in commercial aircraft are desired.

BRIEF DESCRIPTION OF THE DRAWINGS

Features and advantages of the present disclosure will be apparent from the following, more particular, description of various exemplary embodiments, as illustrated in the accompanying drawings, wherein like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 is a schematic perspective view of an aircraft having a gas turbine engine according to an embodiment of the present disclosure.

FIG. 2 is a schematic, cross-sectional view, taken along line 2-2 in FIG. 1 of the gas turbine engine of the aircraft shown in FIG. 1.

FIG. 3 is a perspective view of an unducted single fan engine that may be used with the aircraft shown in FIG. 1.

FIG. 4 is a schematic, cross-sectional view, taken along line 4-4 in FIG. 3, of the unducted single fan engine shown in FIG. 3.

FIG. 5 is a schematic, cross-sectional view, taken along line 2-2 in FIG. 1, of a turbojet engine that may be used with the aircraft shown in FIG. 1.

FIG. 6 is a cross-sectional view of a first combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

FIG. 7 is a cross-sectional view of a second combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

FIG. 8 is a cross-sectional view of a third combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

FIG. 9 is a cross-sectional view of a fourth combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

2

FIG. 10 is a cross-sectional view of a fifth combustor for the gas turbine engine shown in FIG. 2, showing detail 6 of FIG. 2.

FIG. 11 is a graph illustrating combustor length (squared) as a function of combustor height, according to embodiments of the present disclosure.

FIG. 12 is a graph illustrating combustor size rating as a function of core air flow parameter in engine gas turbine engines using hydrogen fuel, according to embodiments of the present disclosure.

FIG. 13 is a cross-sectional view of a fuel nozzle and swirler for use with a gas turbine engine, according to embodiments of the present disclosure.

FIG. 14 is an enlarged perspective view of a cross section of the fuel nozzle of FIG. 13, according to embodiments of the present disclosure.

FIG. 15 is an enlarged cross-sectional view of an outlet of the fuel nozzle of FIG. 13, according to embodiments of the present disclosure.

FIG. 16 is a cross-sectional view of an alternate fuel nozzle and swirler for use with a gas turbine engine, according to embodiments of the present disclosure.

FIG. 17 is a cross-sectional view of an alternative outlet for a fuel nozzle, according to embodiments of the present disclosure.

FIG. 18 is a cross-sectional view of another alternative outlet for a fuel nozzle, according to embodiments of the present disclosure.

FIG. 19 is a cross-sectional view of yet another alternative outlet for a fuel nozzle, according to embodiments of the present disclosure.

FIG. 20 is a cross-sectional view of a convex shape for a fuel nozzle, according to embodiments of the present disclosure.

FIG. 21 is a cross-sectional view of a concave shape for a fuel nozzle, according to embodiments of the present disclosure.

FIGS. 22-33 are plots illustrating the variation in the amount or rate of swirl provided by a fuel nozzle assembly in accordance with exemplary embodiments of the present disclosure.

DETAILED DESCRIPTION

Features, advantages, and embodiments of the present disclosure are set forth or apparent from a consideration of the following detailed description, drawings, and claims. Moreover, it is to be understood that the following detailed description is exemplary and intended to provide further explanation without limiting the scope of the disclosure as claimed.

Various embodiments are discussed in detail below. While specific embodiments are discussed, this is done for illustration purposes only. A person skilled in the relevant art will recognize that other components and configurations may be used without departing from the spirit and scope of the present disclosure.

As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

The terms “forward” and “aft” refer to relative positions within a gas turbine engine or vehicle, and refer to the normal operational attitude of the gas turbine engine or vehicle. For example, with regard to a gas turbine engine, forward refers to a position closer to an engine inlet and aft refers to a position closer to an engine nozzle or exhaust.

The terms “upstream” and “downstream” refer to the relative direction with respect to fluid flow in a fluid pathway. For example, “upstream” refers to the direction from which the fluid flows, and “downstream” refers to the direction to which the fluid flows.

The terms “coupled,” “fixed,” “attached,” “connected,” and the like, refer to both direct coupling, fixing, attaching, or connecting as well as indirect coupling, fixing, attaching, or connecting through one or more intermediate components or features, unless otherwise specified herein.

The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

Approximating language, as used herein throughout the specification and claims, is applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “approximately,” and “substantially” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a one, two, four, ten, fifteen, or twenty percent margin in either individual values, range(s) of values and/or endpoints defining range(s) of values.

The term “bypass ratio,” unless stated otherwise, means the bypass ratio at take off conditions. The term bypass ratio as used herein means the ratio between the mass flow rate of air flow accelerated by the engine that bypasses the engine core to the mass flow rate of the air flow entering the engine core. For example, in an exemplary engine such as the turbofan engine **100** depicted in FIG. **2** and discussed further below, the bypass ratio is the ratio of the mass flow rate of the air flow entering the bypass air flow passage **140** to the mass flow rate of the air flow entering the core air flow path **121**. The bypass ratio can also be estimated as a ratio of the area of an inlet to the bypass duct (e.g., inlet of the bypass air flow passage **140**, discussed below) or an area swept by a rotor (e.g., the area swept by fan blades **322**, discussed below) to the area of the inlet to the engine core (e.g., inlet of the core air flow path **121**).

The term “thrust,” unless stated otherwise, means the maximum thrust at take off. This meaning of thrust is adopted when computing a core airflow parameter (relationship (2), below).

Here and throughout the specification and claims, range limitations are combined, and interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

Combustible hydrocarbon liquid fuel, such as Jet-A fuel, has long been used in gas turbine engines and the components of gas turbine engines, particularly, the combustor, have been designed for such fuels. A hydrogen fuel may be utilized to eliminate carbon dioxide emissions from commercial aircraft. Hydrogen fuel, however, poses a number of challenges as compared to combustible hydrocarbon liquid fuel, such as Jet-A fuel. Hydrogen fuel, for example, is a highly reactive fuel that burns at higher temperatures than combustible hydrocarbon liquid fuel. Hydrogen fuel also has much higher flame speeds. For example, the laminar flame

speed for a hydrogen fuel of diatomic hydrogen is an order of magnitude greater than the laminar flame speed for Jet-A fuel.

When testing hydrogen fuel in current gas turbine engines with rich burn combustors, we, the inventors, observed that the higher combustion temperature of hydrogen fuel results in increased production of nitrogen oxides (“NOx”), as compared to combustible hydrocarbon liquid fuel. We also observed in our testing that NOx emissions are sensitive to combustor residence time. As noted above, hydrogen fuel is highly reactive (relative to other fuels) with a wide range of flammability limits and very high flame speeds, resulting in a very short hydrogen flame close to the front end of the combustor. With such a short flame, the post-flame residence time increases for combustors designed for Jet-A fuel. These findings resulted in a realization that when designing a hydrogen fuel combustor to meet NOx emission targets, the combustor residence time needs to be reduced by more than about fifty percent. To find a suitable combustor design for gas turbine engines using hydrogen fuel, we conceived of a wide variety of combustors having different shapes and sizes in order to determine which embodiment(s) were most promising for a variety of contemplated engine designs and thrust classes. The various embodiments, as described herein and as shown in the figures, are combustors that are sized to meet NOx emissions targets.

FIG. **1** is a perspective view of an aircraft **10** that may implement various preferred embodiments. The aircraft **10** includes a fuselage **12**, wings **14** attached to the fuselage **12**, and an empennage **16**. The aircraft **10** also includes a propulsion system that produces a propulsive thrust required to propel the aircraft **10** in flight, during taxiing operations, and the like. The propulsion system for the aircraft **10** shown in FIG. **1** includes a pair of engines **20**. In this embodiment, each engine **20** is attached to one of the wings **14** by a pylon **18** in an under-wing configuration. Although the engines **20** are shown attached to the wing **14** in an under-wing configuration in FIG. **1**, in other embodiments, the engine **20** may have alternative configurations and be coupled to other portions of the aircraft **10**. For example, the engine **20** may additionally or alternatively include one or more aspects coupled to other parts of the aircraft **10**, such as, for example, the empennage **16** and the fuselage **12**.

As will be described further below with reference to FIG. **2**, the engines **20** shown in FIG. **1** are gas turbine engines that are each capable of selectively generating a propulsive thrust for the aircraft **10**. The amount of propulsive thrust may be controlled at least in part based on a volume of fuel provided to the gas turbine engines **20** via a fuel system **200**. The fuel is stored in a fuel tank **212** of the fuel system **200**. As shown in FIG. **1**, at least a portion of the fuel tank **212** is located in each wing **14** and a portion of the fuel tank **212** is located in the fuselage **12** between the wings **14**. The fuel tank **212**, however, may be located at other suitable locations in the fuselage **12** or the wing **14**. The fuel tank **212** may also be located entirely within the fuselage **12** or the wing **14**. The fuel tank **212** may also be separate tanks instead of a single, unitary body, such as, for example, two tanks each located within a corresponding wing **14**.

Although the aircraft **10** shown in FIG. **1** is an airplane, the embodiments described herein may also be applicable to other aircraft **10**, including, for example, helicopters and unmanned aerial vehicles (UAV). The aircraft discussed herein are fixed-wing aircraft or rotor aircraft that generate lift by aerodynamic forces acting on, for example, a fixed wing (e.g., wing **14**) or a rotary wing (e.g., rotor of a helicopter), and are heavier-than-air aircraft, as opposed to

5

lighter-than-air aircraft (such as a dirigible). The engine 20 may be used in various other applications including stationary power generation systems and other vehicles beyond the aircraft 10 explicitly described herein, such as boats, ships, cars, trucks, and the like.

FIG. 2 is a schematic, cross-sectional view of one of the engines 20 used in the propulsion system for the aircraft 10 shown in FIG. 1. The cross-sectional view of FIG. 2 is taken along line 2-2 in FIG. 1. For the embodiment depicted in FIG. 2, the engine 20 is a high bypass turbofan engine that is referred to as a turbofan engine 100 herein. The turbofan engine 100 has an axial direction A (extending parallel to a longitudinal centerline axis 101, shown for reference in FIG. 2), a radial direction R, and a circumferential direction. The circumferential direction (not depicted in FIG. 2) extends in a direction rotating about the axial direction A. The turbofan engine 100 includes a fan section 102 and a turbomachine 104 disposed downstream from the fan section 102.

The turbomachine 104 depicted in FIG. 2 includes a tubular outer housing or nacelle 106 and an inlet 108. Within the housing 106 there is an engine core, which includes, in a serial flow relationship, a compressor section including a booster or low-pressure (LP) compressor 110 and a high-pressure (HP) compressor 112, a combustion section 150 (also referred to herein as a combustor 150), a turbine section including a high-pressure (HP) turbine 116 and a low-pressure (LP) turbine 118, and a jet exhaust nozzle section 120. The compressor section, the combustor 150, and the turbine section together define at least in part a core air flow path 121 extending from the inlet 108 to the jet exhaust nozzle section 120. The turbofan engine 100 further includes one or more drive shafts. More specifically, the turbofan engine includes a high-pressure (HP) shaft or spool 122 drivably connecting the HP turbine 116 to the HP compressor 112, and a low-pressure (LP) shaft or spool 124 drivably connecting the LP turbine 118 to the LP compressor 110.

The fan section 102 shown in FIG. 2 includes a fan 126 having a plurality of fan blades 128 coupled to a disk 130. The fan blades 128 and the disk 130 are rotatable, together, about the longitudinal centerline axis 101 by the LP shaft 124. The booster 108 may also be directly driven by the LP shaft 124, as depicted in FIG. 2. The disk 130 is covered by a rotatable front hub 132 aerodynamically contoured to promote an air flow through the plurality of fan blades 128. Further, an annular fan casing or outer nacelle 134 is provided, circumferentially surrounding the fan 126 and/or at least a portion of the turbomachine 104. The nacelle 134 is supported relative to the turbomachine 104 by a plurality of circumferentially spaced outlet guide vanes 136. A downstream section 138 of the nacelle 134 extends over an outer portion of the turbomachine 104 so as to define a bypass air flow passage 140 therebetween.

The turbofan engine 100 is operable with the fuel system 200 and receives a flow of fuel from the fuel system 200. As will be described further below, the fuel system 200 includes a fuel delivery assembly 202 providing the fuel flow from the fuel tank 212 to the turbofan engine 100, and, more specifically, to a plurality of fuel nozzles 442 that inject fuel into a combustion chamber 430 of the combustor 150.

The turbofan engine 100 also includes various accessory systems to aid in the operation of the turbofan engine 100 and/or an aircraft including the turbofan engine 100. For example, the turbofan engine 100 may include a main lubrication system 171, a compressor cooling air (CCA) system 173, an active thermal clearance control (ATCC) system 175, and a generator lubrication system 177, each of

6

which is depicted schematically in FIG. 2. The main lubrication system 171 is configured to provide a lubricant to, for example, various bearings and gear meshes in the compressor section, the turbine section, the HP spool 122, and the LP shaft 124. The lubricant provided by the main lubrication system 171 may increase the useful life of such components and may remove a certain amount of heat from such components through the use of one or more heat exchangers. The compressor cooling air (CCA) system 173 provides air from one or both of the HP compressor 112 or LP compressor 110 to one or both of the HP turbine 116 or LP turbine 118. The active thermal clearance control (ATCC) system 175 acts to minimize a clearance between tips of turbine blades and casing walls as casing temperatures vary during a flight mission. The generator lubrication system 177 provides lubrication to an electronic generator (not shown), as well as cooling/heat removal for the electronic generator. The electronic generator may provide electrical power to, for example, a startup electrical motor for the turbofan engine 100 and/or various other electronic components of the turbofan engine 100 and/or an aircraft including the turbofan engine 100.

Heat from these accessory systems 171, 173, 175, and 177, and other accessory systems, may be provided to various heat sinks as waste heat from the turbofan engine 100 during operation, such as to various vaporizers 220, as discussed below. Additionally, the turbofan engine 100 may include one or more heat exchangers 179 within, for example, the turbine section or jet exhaust nozzle section 120 for extracting waste heat from an air flow therethrough to also provide heat to various heat sinks, such as the vaporizers 220, discussed below.

The fuel system 200 of this embodiment is configured to store the fuel for the turbofan engine 100 in the fuel tank 212 and to deliver the fuel to the turbofan engine 100 via the fuel delivery assembly 202. The fuel delivery assembly 202 includes tubes, pipes, and the like, to fluidly connect the various components of the fuel system 200 to the turbofan engine 100. As discussed above, the turbofan engine 100, and, in particular, the combustor 150 discussed herein may be particularly suited for use with hydrogen fuel (diatomic hydrogen). In the embodiments shown in FIG. 2, the fuel is a hydrogen fuel comprising hydrogen, more specifically, diatomic hydrogen. In some embodiments, the hydrogen fuel may consist essentially of hydrogen.

The fuel tank 212 may be configured to hold the hydrogen fuel at least partially in the liquid phase and may be configured to provide hydrogen fuel to the fuel delivery assembly 202 substantially completely in the liquid phase, such as completely in the liquid phase. For example, the fuel tank 212 may have a fixed volume and contain a volume of the hydrogen fuel in the liquid phase (liquid hydrogen fuel). As the fuel tank 212 provides hydrogen fuel to the fuel delivery assembly 202 substantially completely in the liquid phase, the volume of the liquid hydrogen fuel in the fuel tank 212 decreases and the remaining volume in the fuel tank 212 is made up by, for example, hydrogen in the gaseous phase (gaseous hydrogen). As used herein, the term “substantially completely” as used to describe a phase of the hydrogen fuel, refers to at least 99% by mass of the described portion of the hydrogen fuel being in the stated phase, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the hydrogen fuel being in the stated phase.

To store the hydrogen fuel substantially completely in the liquid phase, the hydrogen fuel is stored in the fuel tank 212

at very low (cryogenic) temperatures. For example, the hydrogen fuel may be stored in the fuel tank 212 at about -253 degrees Celsius or less at atmospheric pressure, or at other temperatures and pressures to maintain the hydrogen fuel substantially in the liquid phase. The fuel tank 212 may be made from known materials such as titanium, Inconel®, aluminum, or composite materials. The fuel tank 212 and the fuel system 200 may include a variety of supporting structures and components to facilitate storing the hydrogen fuel in such a manner.

The liquid hydrogen fuel is supplied from the fuel tank 212 to the fuel delivery assembly 202. The fuel delivery assembly 202 may include one or more lines, conduits, etc., configured to carry the hydrogen fuel between the fuel tank 212 and the turbofan engine 100. The fuel delivery assembly 202 thus provides a flow path of the hydrogen fuel from the fuel tank 212 to the turbofan engine 100. The hydrogen fuel is delivered to the engine by the fuel delivery assembly 202 in the gaseous phase, the supercritical phase, or both (e.g., the gaseous phase and the supercritical phase). The fuel system 200 thus includes a vaporizer 220 in fluid communication with the fuel delivery assembly 202 to heat the liquid hydrogen fuel flowing through the fuel delivery assembly 202. The vaporizer 220 is positioned in the flow path of the hydrogen fuel between the fuel tank 212 and the turbofan engine 100. The vaporizer 220 may be positioned at least partially within the fuselage 12 or the wing 14 (both shown in FIG. 1), such as at least partially within the wing 14. The vaporizer 220 may, however, be positioned at other suitable locations in the flow path of the hydrogen between the fuel tank 212 and the turbofan engine 100. For example, the vaporizer 220 may be positioned external to the fuselage 12 and the wing 14 (both shown in FIG. 1) and positioned at least partially within the pylon 18 (FIG. 1) or the turbofan engine 100 (FIG. 2). When positioned in the turbofan engine 100, the vaporizer may be located in the nacelle 134, for example. Although only one vaporizer 220 is shown in FIG. 2, the fuel system 200 may include multiple vaporizers 220. For example, when a vaporizer 220 is positioned in the turbofan engine 100 or in the pylon 18 and functions as a primary vaporizer configured to operate once the turbofan engine 100 is in a thermally stable condition, another vaporizer 220 is positioned upstream of the primary vaporizer and proximate to the fuel tank 212, and functions as a primer vaporizer during start-up (or prior to start-up) of the turbofan engine 100.

The vaporizer 220 is in thermal communication with at least one heat source 222, 224. In this embodiment, the vaporizer 220 is in thermal communication with a primary heat source 222 and an auxiliary heat source 224. In this embodiment, primary heat source 222 is waste heat from the turbofan engine 100, and the vaporizer 220 is, thus, thermally connected to at least one of the main lubrication system 171, the compressor cooling air (CCA) system 173, the active thermal clearance control (ATCC) system 175, the generator lubrication system 177, and the heat exchangers 179 to extract waste heat from the turbofan engine 100 to heat the hydrogen fuel. In such a manner, the vaporizer 220 is configured to operate by drawing heat from the primary heat source 222 once the turbofan engine 100 is capable of providing enough heat, via the auxiliary heat source 224, to the vaporizer 220, in order to facilitate operation of the vaporizer 220.

The vaporizer 220 may be heated by any suitable heat source, and, in this embodiment, for example, the auxiliary heat source 224 is a heat source external to the turbofan engine 100. The auxiliary heat source 224 may include, for

example, an electrical power source, a catalytic heater or burner, and/or a bleed air flow from an auxiliary power unit. The auxiliary heat source 224 may be integral to the vaporizer 220, such as when the vaporizer 220 includes one or more electrical resistance heaters, or the like, that are powered by the electrical power source. In this configuration the auxiliary heat source 224 may provide heat for the vaporizer 220 independent of whether or not the turbofan engine 100 is running and can be used, for example, during start-up (or prior to start-up) of the turbofan engine 100.

As noted, the vaporizer 220 is in communication with the flow of the hydrogen fuel through the fuel delivery assembly 202. The vaporizer 220 is configured to draw heat from at least one of the primary heat source 222 and the auxiliary heat source 224 to heat the flow of hydrogen fuel from a substantially completely liquid phase to a substantially completely gaseous phase or to a substantially completely supercritical phase.

The fuel system 200 also includes a high-pressure pump 232 in fluid communication with the fuel delivery assembly 202 to induce the flow of the hydrogen fuel through the fuel delivery assembly 202 to the turbofan engine 100. The high-pressure pump 232 may generally be the primary source of pressure rise in the fuel delivery assembly 202 between the fuel tank 212 and the turbofan engine 100. The high-pressure pump 232 may be configured to increase a pressure in the fuel delivery assembly 202 to a pressure greater than a pressure within the combustion chamber 430 of the combustor 150 of the turbofan engine 100, and to overcome any pressure drop of the components placed downstream of the high-pressure pump 232.

The high-pressure pump 232 is positioned within the flow of hydrogen fuel in the fuel delivery assembly 202 at a location downstream of the vaporizer 220. In this embodiment, the high-pressure pump 232 is positioned external to the fuselage 12 and the wing 14, and is positioned at least partially within the pylon 18, or at least partially within the turbofan engine 100. More specifically, the high-pressure pump 232 is positioned within the turbofan engine 100. With the high-pressure pump 232 located in such a position, the high-pressure pump 232 may be any suitable pump configured to receive the flow of hydrogen fuel in substantially completely a gaseous phase or a supercritical phase. In other embodiments, however, the high-pressure pump 232 may be positioned at other suitable locations, including other positions within the flow path of the hydrogen fuel. For example, the high-pressure pump 232 may be located upstream of the vaporizer 220 and may be configured to receive the flow of hydrogen fuel through the fuel delivery assembly 202 in a substantially completely liquid phase.

The fuel system 200 also includes a metering unit in fluid communication with the fuel delivery assembly 202. Any suitable metering unit may be used including, for example, a fuel metering valve 234 placed in fluid communication with the fuel delivery assembly 202. The fuel delivery assembly 202 is configured to provide the fuel metering valve 234, and the fuel metering valve 234 is configured to receive hydrogen fuel. In this embodiment, the fuel metering valve 234 is positioned downstream of the high-pressure pump 232. The fuel metering valve 234 is further configured to provide the flow of the hydrogen fuel to the turbofan engine 100 in a desired manner. The fuel metering valve 234 is configured to provide a desired volume of the fuel at, for example, a desired flow rate, to a fuel manifold 236 of the turbofan engine 100. The fuel manifold 236 then distributes (provides) the hydrogen fuel received to a plurality of fuel nozzles 442 (see FIG. 6) within the combustion section 150

of the turbofan engine **100** where the hydrogen fuel is mixed with compressed air, and the mixture of hydrogen fuel and compressed air is combusted to generate combustion gases that drive the turbofan engine **100**. Adjusting the fuel metering valve **234** changes the volume of fuel provided to the combustion chamber **430** (see FIG. **6**) of the combustor **150** and, thus, changes the amount of propulsive thrust produced by the turbofan engine **100** to propel the aircraft **10**.

Although the turbofan engine **100** is shown as a direct drive, fixed-pitch turbofan engine **100**, in other embodiments, a gas turbine engine may be a geared gas turbine engine (i.e., including a gearbox between the fan **126** and shaft driving the fan, such as the LP shaft **124**), may be a variable pitch gas turbine engine (i.e., including a fan **126** having a plurality of fan blades **128** rotatable about their respective pitch axes), etc. Additionally, in still other exemplary embodiments, the exemplary turbofan engine **100** may include or be operably connected to any other suitable accessory systems. Additionally, or alternatively, the exemplary turbofan engine **100** may not include or be operably connected to one or more of the accessory systems **171**, **173**, **175**, and **177**, discussed above.

The turbofan engine **100** discussed herein is an example of the engine **20** in which the combustors **150** discussed herein may be used. In other embodiments, other suitable engines may be utilized with aspects of the present disclosure. For example, FIGS. **3** and **4** show an unducted single fan (USF) engine **300** that may be used as the engine **20** of the aircraft **10** and implement the fuel system described above, and combustor designs discussed further below. FIG. **3** is a perspective view of the USF engine **300** and FIG. **4** is a cross-sectional view taken along a line **4-4** in FIG. **3**.

The USF engine **300** includes a housing **302**. The housing **302** may be formed of a nacelle **310** and spinner **320**. The nacelle **310** and/or the spinner **320** house internal components of the USF engine **300**. For example, the nacelle **310** houses a torque producing system **312** coupled to a shaft **314**. The torque producing system **312** in the embodiments discussed herein is a gas turbine engine, such as the turbomachine **104** discussed above with reference to FIG. **2** and, thus, the nacelle **310** of this embodiment is similar to the tubular outer housing **106** discussed above. As the turbomachine **104** used as the torque producing system **312** of the USF engine has the same or similar components and features as the turbomachine **104** discussed above, a detailed description of the components of the turbomachine **104** used in of the USF engine **300** is omitted.

The torque producing system **312** and the shaft **314** are configured to operate (e.g., to rotate) the spinner **320**. One or more fan blades **322** are coupled to the spinner **320**. More specifically, the spinner **320** includes a fan hub **324**, and the fan blades **322** are coupled to the fan hub **324**. The spinner **320** rotates with respect to the nacelle **310**. Coupled to the nacelle **310** may be one or more outlet guide vanes **326**. In this embodiment, the outlet guide vanes **326** are positioned aft of the fan blades **322**. During operation, the one or more fan blades **322** (by virtue of the connection to the spinner **320**) rotate circumferentially around a longitudinal centerline **304**, in this embodiment, and the nacelle **310** is stationary such that the one or more outlet guide vanes **326** do not rotate around the longitudinal centerline **304** and are, thus, stationary with respect to rotation about the longitudinal centerline **304**. Although the outlet guide vanes **326** are stationary with respect to the longitudinal centerline **304**, the

outlet guide vanes **326** are capable of being rotated or moved with respect to the nacelle **310**, for example, in the direction **A** of FIG. **4**.

During operation of the USF engine **300**, air flows from the left side of FIG. **4** toward the right side of FIG. **4**. A portion of the air flow may flow past the fan blades **322** and the outlet guide vanes **326**. A portion of the air flow may enter the nacelle **310** through the annular inlet **108** to be mixed with the hydrogen fuel for combustion in a combustor **150** of the USF engine **300** and exit through an outlet **120**. The outlet guide vanes **326** may be movable with respect to the nacelle **310** to guide the air flow in a particular direction. Each outlet guide vane **326** may be movable to adjust the lean, pitch, sweep, or any combination thereof, of the outlet guide vane **326**.

In the embodiment shown in FIGS. **3** and **4**, a forward end or front portion of the housing **302** includes the one or more fan blades **322** and the one or more outlet guide vanes **326**. In other embodiments, the one or more fan blades **322** and the one or more outlet guide vanes **326** may have a different arrangement with respect to the housing **302**. For example, the one or more fan blades **322** and the one or more outlet guide vanes **326** may be located on an aft end or rear portion of the housing **302**, such as coupled to a rear portion of the housing **302**.

In other embodiments, an engine according to the disclosure may be configured to have either the stationary vanes positioned forward of the rotating blades **322** (thus, the blades **326** are inlet guide vanes) or both the blades **326** and blades **322** configured to operate in a counter-rotating fashion. Either “pusher” or “puller” configurations are contemplated. In each of these alternative embodiments, the fuel delivery system **200** and combustor **150**, as described in great detail below, may be used. An example of a suitable engine configuration for a counter-rotating engine is shown and described in FIG. **1** and col. **3**, line **43** through col. **4**, line **11** of U.S. Pat. No. **10,800,512**, hereby incorporated by reference for all purposes. Alternative embodiments of the USF engine **300** are shown and described in FIGS. **6**, **7**, and **8** and col. **4**, line **51** through col. **5**, line **19** of U.S. Pat. No. **10,704,410**, hereby incorporated by reference for all purposes.

In further embodiments, a turbojet engine **350** may be used as the engine **20**. FIG. **5** is a schematic, cross-sectional view of the turbojet engine **350**. The cross-sectional view of FIG. **5** is similar to FIG. **2**, which is taken along line **2-2** in FIG. **1**. The turbojet engine **350** includes the same or similar components of the turbomachine **104** of the turbofan engine **100** and a detailed description of these components is omitted. An exemplary turbojet engine **350** may not include a fan with bypass duct. An exemplary turbojet engine **350** may have high velocity exhaust from the engine, which produces a majority of the thrust for the turbojet engine **350**. In still further embodiments, other suitable gas turbine engines, such as a turboshaft engine, a turboprop engine, and the like, may be utilized with aspects of the present disclosure.

As noted above, we conceived of a wide variety of combustors having different shapes and sizes. FIGS. **6** to **10** show various combustor shapes that can suitably be used as the combustor **150** for the gas turbine engines **20** discussed herein. FIGS. **6** to **10** are a detail views showing detail **6** in FIG. **2**, and, as FIG. **2** is a cross-sectional view, FIGS. **6** to **10** are also cross-sectional views. FIG. **6** shows a first combustor **401**. FIG. **7** shows a second combustor **403**. FIG. **8** shows a third combustor **405**. FIG. **9** shows a fourth combustor **407**. FIG. **10** shows a fifth combustor **409**.

Although the shapes of these combustors **401**, **403**, **405**, **407**, **409** differ, each of these combustors **401**, **403**, **405**, **407**, **409** has similar components, and common reference numerals are used in FIGS. **6** to **10** to for the same or similar components of these combustors **401**, **403**, **405**, **407**, **409**. Accordingly, the following detailed description of the first combustor **401** also applies to the second combustor **403**, the third combustor **405**, the fourth combustor **407**, and the fifth combustor **409**. Some components, such as the combustor casing **410**, for example, may not be shown in each figure, but such components may nevertheless be applicable to the combustors **403**, **405**, **407**, **409**.

As shown in FIG. **6**, combustor **401** includes a combustor casing **410** and a combustor liner **420**. The combustor casing **410** of this embodiment has an outer casing **412** and an inner casing **414**, and the combustor liner **420** of this embodiment has an outer liner **422** and an inner liner **424**. A combustion chamber **430** is formed within the combustor liner **420**. More specifically, the outer liner **422** and the inner liner **424** are disposed between the outer casing **412** and the inner casing **414**. The outer liner **422** and the inner liner **424** are spaced radially from each other such that the combustion chamber **430** is defined therebetween. The outer casing **412** and the outer liner **422** form an outer passage **416** therebetween, and the inner casing **414** and the inner liner **424** form an inner passage **418** therebetween. In this embodiment, the combustor **401** is a single annular combustor, but, in other embodiments, the combustor **401** may be any other combustor, including, but not limited to a double annular combustor.

The combustion chamber **430** has a forward end **432** (downstream end) and an aft end **434** (upstream end). The fuel nozzle **442** is positioned at the forward end **432** of the combustion chamber **430**. The fuel nozzle **442** of this embodiment is part of a swirler/fuel nozzle assembly **440**. In this embodiment, when the combustor **401** is an annular combustor **150**, a plurality of fuel nozzles **442** is arranged in an annular configuration with the plurality of fuel nozzles **442** (the swirler/fuel nozzle assemblies **440**) aligned in a circumferential direction of the combustor **401**.

As discussed above, the compressor section, the combustor **401**, and the turbine section form, at least in part, the core air flow path **121** extending from the annular inlet **108** to the jet exhaust nozzle section **120**. Air entering through the annular inlet **108** is compressed by blades of a plurality of fans of the LP compressor **110** and HP compressor **112**. A cowl assembly **450** is coupled to the upstream ends of outer liner **422** and the inner liner **424**, respectively. An annular opening **452** formed in the cowl assembly **450** enables compressed air from the compressor section (indicated by arrow **B**) to enter the combustor **401**. The compressed air flows through the annular opening **452** to support combustion. Another portion of the compressed air flows around the outside of the combustor liner **420** through the outer passage **416** and the inner passage **418**. This air is introduced into the combustion chamber **430** through a plurality of circumferentially spaced dilution holes **426** formed in the combustor liner **420** at positions downstream of the fuel nozzle **442**.

An annular dome plate **454** extends between, and is coupled to, outer liner **422** and the inner liner **424** near their upstream ends. The plurality of circumferentially spaced swirler/fuel nozzle assemblies **440** is coupled to dome plate **454**. Each swirler/fuel nozzle assembly **440** receives compressed air from the annular opening **452**. The swirler/fuel nozzle assembly **440** includes a swirler **444** that is used to generate turbulence in the air. The fuel nozzle **442** injects fuel into the turbulent air flow and the turbulence promotes

rapid mixing of the fuel with the air. The resulting mixture of fuel and compressed air is discharged into combustion chamber **430** and combusted in the combustion chamber **430**, generating combustion gases (combustion products), which accelerate as the combustion gases leave the combustion chamber **430**.

A turbine nozzle **460** is disposed at the outlet of the combustion chamber **430**. The turbine nozzle **460** may be a stage **1** turbine nozzle. The turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424** at the downstream (aft) ends of each of the outer liner **422** and the inner liner **424**. The turbine nozzle **460** of this embodiment includes an outer band **462** and an inner band **464** coupled to outer liner **422** and the inner liner **424**, respectively. The turbine nozzle **460** also includes a leading edge **466**, which in this embodiment is the location where the turbine nozzle **460** is coupled to outer liner **422** and the inner liner **424**, and the outer band **462** and the inner band **464** each has the leading edge **466**. The turbine nozzle **460** further includes a plurality of circumferentially spaced vanes **468** extending between the outer band **462** and the inner band **464**. The vanes **468** extend in a generally radial direction. The vanes **468**, and the turbine nozzle **460**, is a static component and the vanes **468** may be cured to direct (e.g., spin or swirl) the combustion gases to turn the turbines (e.g., drive the turbine blades) of the first stage of the HP turbine **116**. In this embodiment, the turbine section is a multi-stage turbine and these combustion gases will drive subsequent stages of the HP turbine **116** and the LP turbine **118**. The turbine nozzle **460** may, thus, also be referred to as a stage one nozzle (SIN). As discussed above the HP turbine **116** and the LP turbine **118**, among other things, drive the LP compressor **110** and HP compressor **112**.

As noted above, we realized that when designing hydrogen fuel combustor to meet NOx emission targets, the combustor residence time needs to be reduced. We sized the combustor **401**, and more specifically, the combustor liner **420** for various gas turbine engines and flow rates. These different embodiments are shown below in Table 1 and were developed for different bypass ratios and thrust classes of engines, characterized by the core airflow. In particular, we considered the height **H**, also referred to as burner dome height, of the combustion chamber **430** and the length **L**, also referred to as burner length, of the combustion chamber. Diluents could be used to suppress the temperature, and, thus, NOx production, in the combustion chamber **430** when hydrogen is used as the fuel. With the combustor sized as described in these embodiments, hydrogen fuel can be used without the need of diluents. In some embodiments, no diluent is added to the combustion chamber **430** and the fuel is substantially completely diatomic hydrogen without diluent. As used herein, the term "substantially completely," as used to describe the amount of a particular element or molecule (e.g., diatomic hydrogen), refers to at least 99% by mass of the described portion of the element or molecule, such as at least 97.5%, such as at least 95%, such as at least 92.5%, such as at least 90%, such as at least 85%, or such as at least 75% by mass of the described portion of the element or molecule.

FIGS. **6** to **10** illustrate how the height **H** and length **L** may be determined for the different shapes of combustion liners **420** shown in these figures. The height **H** of the combustion chamber **430** is taken at the forward end **432** of the combustion chamber **430**. The height **H** is the maximum height between an inner surface of the outer liner **422** and an inner surface of the inner liner **424** at the forward end **432** of the combustion chamber **430**. The height **H** is measured along a

13

line (referred to as a forward line **472**, herein) that is generally orthogonal the inner surfaces of the outer liner **422** and the inner liner **424**. The forward line **472** may be orthogonal to a central axis **477** of the fuel nozzle assembly **440** and/or the fuel nozzle **442**. In this manner, the height H may be orthogonal to the central axis **477**. In some embodiments, the height H measured using the forward line is the maximum height of the combustion chamber **430** and may also be the maximum dome height of the combustion chamber **430**.

The length L of the combustion chamber **430** is the distance between forward line **472** and the leading edge **466** of the turbine nozzle **460**. As with the height H, a line (referred to as the aft line **474**, herein) can be drawn from the leading edge **466** at the outer liner **422** and leading edge at the inner liner **424**. Each of the forward line **472** and the aft line **474** has a midpoint (midpoint **476** and midpoint **478**, respectively) that is halfway between the outer liner **422** and the inner liner **424**. The length L can be measured from the midpoint **476** of the forward line **472** to the midpoint **478** of the aft line **474**. The midpoint **478** may be the midspan height of the turbine nozzle **460**.

When developing a gas turbine engine, the interplay between components can make it particularly difficult to select or to develop one component during engine design and prototype testing, especially, when some components are at different stages of completion. For example, one or more components may be nearly complete, yet one or more other components may be in an initial or preliminary phase such that only one (or a few) design parameters are known. It is desired to arrive at what is possible at an early stage of design, so that the down selection of candidate optimal designs, given the tradeoffs, become more possible. Here-
 tofore, the process has sometimes been more ad hoc, selecting one design or another without knowing the impact when a concept is first taken into consideration. For example, various aspects of the fan **126** design, the HP compressor **112** design, and/or the LP compressor **110** design may not be known, but such components impact the core air flow through the core air flow path **121**, and, thus, may influence the design of the combustion chamber **430**.

We desire to narrow the range of configurations or combination of features that can yield favorable results given the constraints of the design, feasibility, manufacturing, certification requirements, etc., early in the design selection process to avoid wasted time and effort. During the course of the evaluation of different embodiments as set forth above, we, the inventors, discovered, unexpectedly, that there exists a relationship between the burner length and the burner dome height, which uniquely identifies a finite and readily ascertainable (in view of this disclosure) number of embodiments suitable for a particular architecture that can meet NOx emissions for hydrogen fuel and provide desired flame residence times. This relationship is referred to by the inventors as the combustor size rating (CSR) (in), and is defined according to the following relationship (1) between burner length L (in) and burner dome height H (in):

$$\text{Combustor Size Rating (CSR)} = (L)^2/(H) \quad (1)$$

As discussed further below, we have identified a range of the Combustor Size Ratings that enable a combustion chamber **430** to be designed for a gas turbine engine **20** using hydrogen fuel. This relationship is applicable over a wide range of thrust class and engine designs. Using this unique

14

relationship, a combustor **150** design can be developed early in the design process that meets NOx emissions targets and reduces engine weight for gas turbine engines using hydrogen fuel.

Table 1 describes exemplary embodiments 1 to 24 identifying the CSR for various hydrogen fuel burning engines. The embodiments 1 to 24 may be engines with either rich burn combustors or lean burn combustors. Each of embodiments 1 to 24 burns hydrogen fuel. Embodiments 1 to 24 may represent any of the engines described with respect to FIGS. **1** to **5** and can be applied to any of the combustion chamber **430** shapes shown in FIGS. **6** to **10**. In Table 1, the CSR is determined based on the relationship (1) described above. A core air flow parameter (CAFP) (kN) is defined according to the following relationship (2) between thrust (kN) and bypass ratio, both at take off.

$$\text{Core Air Flow Parameter} = \frac{\text{Thrust}}{\text{Bypass Ratio}} \quad (2)$$

The burner length is the length L identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two inches and six inches. In embodiments 1 to 24, the burner length squared may be between six square inches and thirty-five square inches. The burner dome height is the height H identified with respect to FIGS. **6** to **10**, and in the embodiments 1 to 24 is between two and one half inches and six inches.

TABLE 1

Embodiment	Combustor Size Rating (in)	Core Air Flow Parameter (kN)	Thrust (kN)	Bypass Ratio
1	4.30	38.16	332.39	8.71
2	6.67	49.85	254.26	5.10
3	6.67	53.44	272.53	5.10
4	6.67	51.18	261.03	5.10
5	6.67	52.36	267.03	5.10
6	4.69	21.07	120.10	5.70
7	4.69	23.80	121.40	5.10
8	3.01	12.58	64.53	5.13
9	3.01	12.18	62.49	5.13
10	3.00	16.44	83.70	5.09
11	3.00	15.20	82.10	5.40
12	3.00	14.27	84.20	5.90
13	3.00	14.27	84.20	5.90
14	3.00	14.27	84.20	5.90
15	2.12	36.55	321.60	8.80
16	2.12	39.23	345.20	8.80
17	2.12	40.65	349.20	8.59
18	2.12	40.65	349.20	8.59
19	2.12	37.34	299.81	8.03
20	1.67	13.63	143.10	10.50
21	1.94	51.51	489.30	9.50
22	5.51	40.89	363.90	8.90
23	2.46	12.72	147.28	11.58
24	2.70	5.00	150.00	30.00

The length L may be between 2.63 inches and 5.60 inches. The length L may be between two inches and three inches. The length L may be between two and one half inches and three and one half inches. The height H may be between 2.80 inches and 5.60 inches. The height H may be between two and one half inches and six inches. The height H may be between two and one half inches and five inches. The height H may be between four inches and five inches. The burner length squared may be between 6.89 inches and 31.36 inches. The burner length squared may be between six

15

square inches and thirty-five square inches. The burner length squared may be between six square inches and twenty square inches. The burner length squared may be between six square inches and twelve square inches. The burner length squared may be between eight square inches and twelve square inches. The burner length squared and the height may be any values such that the CSR is less than seven inches. The burner length squared and the height may be any values such that the CSR is less than six inches.

FIG. 11 represents, in graph form, the burner length, squared, as a function of the burner dome height. FIG. 11 shows that the burner length, squared, may be changed based on the burner dome height. An area 500 may present the boundaries of burner length, squared, as a function of burner dome height in which a particular combustor is designed. FIG. 12 represents, in graph form, the CSR as a function of core air flow parameter. Table 1 and FIG. 12 show that CSR may be changed based on a thrust class, as characterized by the core air flow parameter, of an engine. An area 600 may present the boundaries of CSR as a function of the core air flow parameter in which a particular combustor is designed.

As shown in FIG. 12, the CSR is less than seven inches for every core air flow. That is, the CSR is less than seven inches for every thrust class of engine. The CSR may be between 1.67 inches and 6.67 inches. The CSR may be between one inch and seven inches. The CSR may be between one and one half inches and seven inches. The CSR may be between two inches and seven inches. The CSR may be between two inches and six inches. The CSR may be between one inches and five inches. The CSR may be between two inches and five inches. The CSR may be between three inches and five inches. The core air flow parameter may be less than sixty kN. The core air flow parameter may be between five kN and 53.44 kN. The core air flow parameter may be between two and one half kN and sixty kN. The core air flow parameter may be between ten kN and twenty kN. The core air flow parameter may be between thirty kN and forty-five kN.

With continued reference to FIG. 12, the CSR may be a function of the core air flow parameter. The CSR may be based on a thrust of the gas turbine engine. The CSR may be between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN. The CSR may be between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN. The thrust may be between sixty kN and five hundred kN. The thrust may be between 62.49 kN and 489.30 kN. The CSR is defined by a relationship of the burner length, squared, and the burner dome height.

In an extension of the concepts disclosed hereinabove, also provided herein are combustors that include a fuel nozzle and swirler. It has been determined that hydrogen fuel results in higher combustion temperatures as compared to combustible hydrocarbon liquid fuel and that NOx emissions are sensitive to factors including combustor residence times. The present disclosure allows for fuel nozzle assembly structures configured for use with such higher combustion temperatures. More specifically, the use of hydrogen fuel requires greater flame control in order to prevent flame holding or flashback.

The swirler can create a swirling airflow that mixes with fuel. Varying the rate of swirl for the fuel flow from the fuel nozzle and the air flow from the swirler can be utilized to develop complex velocity profiles, which can be tailored to mitigate flame holding, flashback, and recirculation at various radial positions for the fuel nozzle assembly, which can

16

provide for the use of faster-burning or higher-temperature fuels, such as hydrogen or hydrogen mixes. It has further been determined that no swirl within the radial center of the fuel nozzle provides for positioning a recirculation bubble in the aft direction, which can be a recirculating flow of fuel resultant of the wake generated by the fuel, which mitigates flame holding at the fuel nozzle. Increasing the rate of swirl for the fuel from no swirl to matching the swirl at the radial interior of the swirler provides for reducing shear between the two flows, as well as eliminating flame holding. Finally, increasing the tangential swirl in a radially outward direction within the swirler provides for mitigating flame holding on the flare cone.

It is desirable to have a number of openings in the fuel nozzle and the numbering, size, orientation, and placement of the openings can be varied to achieve the desired velocity distribution at the fuel nozzle exit. Furthermore, the openings can vary in size to vary the fuel momentum profile at the nozzle exit. Such velocity distribution and momentum profiles can be tailored to the particular fuel nozzle or engine to mitigate recirculation, flame holding, or flashback. In addition to resulting in reduced residence times, the combustors described herein including the described fuel nozzle and swirler further permit an increase in fuel efficiency and allows for NOx emissions to be met.

FIG. 13 illustrates a fuel nozzle assembly 700 including a fuel nozzle 702 and a swirler 704 provided annularly about the fuel nozzle 702. A dome (not shown) may be provided forward of and adjacent to the swirler 704. A nozzle supply passage 706, a fuel nozzle cap 708, and a nozzle outlet 710 can be included in the fuel nozzle 702. A set of openings 740 can be included in the nozzle cap 708, the set of openings 740 permitting fuel to exhaust from the nozzle supply passage 706. The swirler 704 can include a swirler supply passage 720 at least partially circumscribing the fuel nozzle 702 and exhausting to a flare cone 722. A set of swirler vanes 724 are provided within the swirler supply passage 720 to impart a tangential or helical swirl to the air provided by the swirler 704.

A splitter 726 extends from the swirler vanes 724 to separate airflow within the swirler supply passage 720 into two swirling air streams contained within an outer diameter passage 728 and an inner diameter passage 730 defined by the splitter 726. The splitter 726 extends axially to a downstream end 732, such that the downstream end 732 is coaxial with the fuel nozzle supply passage 706 or the longitudinal centerline axis 101 FIG. 2, while it is contemplated that an angular offset exists for the downstream end 732 relative to the fuel nozzle supply passage 706 or relative to a longitudinal axis 712 defined by the fuel nozzle 702, the swirler 704, or a longitudinal axis defined by the combustion section 150 (FIG. 2) containing the fuel nozzle assembly 700. Such an offset can be used to impart a directionality to the combusted fuel, relative to the longitudinal extent of the fuel nozzle 702. Additionally, the fuel nozzle 702 can be cylindrical, such that a radial axis 714 can be defined perpendicular to the longitudinal axis 712 although this need not be the case.

FIG. 14 illustrates a portion of the fuel nozzle 702 including the set of openings 740 provided in the fuel nozzle cap 708. The set of openings 740 can be arranged, by way of non-limiting example, in circumferential rows of openings 742a-e, which are defined circumferentially relative to an axial extent of the fuel nozzle supply passage 706 or the longitudinal axis 712. It will be understood that the set of openings 740 can have any suitable alternative arrangements.

17

Each opening of the set of openings 740 can include a tangential component, having a centerline 738 arranged as a tangential angle 750, such that a swirl or helical component is applied to fuel supplied through the fuel nozzle cap 708. The tangential angle 750 can be defined relative to the longitudinal axis 712, an axis parallel to the longitudinal axis 712, or to a radius extending from the longitudinal axis 712. Additionally, the tangential angle 750 of each opening of the set of openings 740, or each row defined by the set of openings 742a-e, can be predetermined. More specifically, the tangential component for the set of openings 740 or a row defined by the set of openings 742e near the outer diameter of the fuel nozzle cap 708 can be configured to impart an increased swirl relative to the set of openings 740 or rows defined by the set of openings 742a-d, which are interior of the outer diameter row defined by the set of openings 740e, but utilizing an increasing tangential angle 750 to impart increasing swirl.

By way of non-limiting example, the tangential angle 750 of the set of openings 740 can increase in a direction extending radially outward from the longitudinal axis 712. In such an instance, the tangential angle 750 for each row defined by the set of openings 742a-d can increase the further from the longitudinal axis 712 that the rows defined by the set of openings 742a-d are positioned. In another non-limiting example, a center or central opening 742a relative to the fuel nozzle cap 708 can also have no tangential component, such as being coaxial with the fuel nozzle supply passage 706. This can push central recirculation from the swirler further aft, which can further mitigate flame holding or flashback. In examples where the set of openings 740 are not arranged in rows, the tangential angle of each opening of the set of openings 740 can increase in a direction extending radially outward from a center of the fuel nozzle cap 708, such that openings of the set of openings 740 further from the longitudinal axis 712 have a greater tangential angle 750 than openings of the set of openings 740 nearer to the longitudinal axis 712.

In another non-limiting example, the openings of the set of openings 740 nearer to the center of the fuel nozzle cap 708 can be smaller, such as having a smaller cross-sectional area, relative to the set of openings 740 nearer to the outer diameter of the fuel nozzle cap 708. This can provide for mitigate jet flapping. Different cross-sectional profiles for one or more openings of the set of openings 740 are further contemplated, such as circular, slot, oval, elliptical, or racetrack in non-limiting examples. It is further contemplated that the shapes can vary based upon the radial relation to the center of the fuel nozzle cap 708, such that the shape varies based upon radial position.

FIG. 15 further illustrates a portion of fuel nozzle lip 744 provided at the nozzle outlet 710 downstream of the fuel nozzle cap 708. The fuel nozzle lip 744 includes an axial portion 746 and a diverging portion 748 extending from the axial portion 746. The axial portion 746 defines a constant cross-sectional area downstream of the fuel nozzle cap 708 and can be coaxial with the fuel nozzle 702. The diverging portion 748 defines an increasing or diverging cross-sectional area. In an alternative example, the nozzle outlet 710 can be formed as converging, diverging, constant, or any combination thereof, which can be defined by linear, curved, or discrete wall geometries defining the nozzle outlet 710, or combinations thereof. The radiused tip of the fuel nozzle lip 744 provides for reducing flow recirculation at the end of the nozzle outlet 710, which can eliminate stagnation points and flame holding.

18

Referring to FIGS. 13-15 simultaneously, in operation, a supply of air is provided via the swirler 704 and a supply of fuel is provided via the fuel nozzle supply passage 706. The supply of air provided via the swirler 704 is imparted with a tangential, swirling, or helical component by the swirler vanes 724 and separated into two flows by the splitter 726. The two flows are formed as an inner diameter flow and an outer diameter flow, which are relative to a radius extending from the longitudinal axis 712 of the fuel nozzle supply passage 706. The two flows separated by the splitter 726 can swirl in the same direction to minimize any shear between the inner diameter flow and the outer diameter flow, while counter-flow is contemplated where increased mixing or turbulence may be advantageous.

Fuel provided from the fuel nozzle 702 is provided with a tangential or swirling component via the set of openings 740 in the fuel nozzle cap 708. The direction of the swirl imparted by the fuel nozzle 702 can be the same direction as that of the air provided by the swirler 704 to reduce or avoid any shear defined between the fuel and the air. In one example, the tangential component for the set of openings 740 can be related to the tangential component imparted by the swirler 704. For example, the tangential angles among the set of openings 740 and the swirler 704 can be complementary, in order to further reduce shear among the different flows. Reduction of the shear reduces the shear layer deficit between the airflow and the fuel flow. The reduction of this deficit can provide for improved distribution for the radial velocity profile, which can provide for maintaining a high axial velocity for both the fuel and the airflow. The high axial velocity mitigates the occurrence of flame holding and flashback on the fuel nozzle assembly 700, permitting the use of higher temperature fuels, such as hydrogen fuels, which can reduce or eliminate carbon emissions. Additionally, the co-swirling of the air and the fuel deters mixing of the air and fuel which provides for greater mixing control, as well as the mitigation of flashback and flame holding under high-temperature operations.

Alternatively, a counter-flow is contemplated to increase mixing between the air and the fuel. The axial portion 746 provides for improved consistency for the velocity profile to maintain a high axial velocity, which can prevent flame holding on the fuel nozzle outlet 710. As the fuel is emitted from the nozzle outlet 710, the diverging portion 748 at the fuel nozzle lip 744 permits the fuel flow to expand which reduces or eliminates low velocity regions on the nozzle lip 744, which mitigates flame holding on the fuel nozzle 702 or swirler 704 hardware.

The splitter 726 provides for improved velocity distribution and control thereof, prior to introduction of the air flow to the fuel flow. The improved velocity distribution provides for preventing flame holding on the fuel nozzle 702, swirler 704, or flare cone 722. The constant area along the splitter 726 that is coaxial with the fuel nozzle supply passage 706 provides for improved flow development for the swirling airflow, which reduces or eliminates the occurrence of a low velocity region created by the splitter 726. The fuel is introduced downstream of the swirler 704 to prevent flame holding against the swirler 704. Similarly, the swirl created by the tangential openings 740 on the fuel nozzle cap 708 can reduce or avoid the occurrence of low velocity on the fuel nozzle outer diameter, which can reduce the opportunity for flame holding or flashback on the fuel nozzle 702.

The fuel nozzle lip 744 is arranged downstream of the fuel nozzle cap 708 to reduce the mixture of air and fuel before the fuel nozzle lip 744, and further reduces the generation of recirculation zones in the region where the air and the fuel

streams mix. The fuel nozzle cap **708** provides for imparting tangential swirl to the fuel supply, while the axial portion **746** and the diverging portion **748** provide a well-defined velocity profile for the fuel supply before the fuel is introduced to the air from the swirler **704**, while reducing or eliminating recirculation zones.

The fuel nozzle examples described herein provide for improved fuel supply to a combustor such as the combustors described above. The improved fuel supply mitigates flame holding or flashback at the fuel nozzle assembly **700**. This in turn allows for the use of higher temperature fuels, such as hydrogen fuels, which provide for improving or maintaining efficiency while reducing or eliminating emissions.

FIG. **16** illustrates a fuel nozzle assembly **800** having a central passage **802** provided within a fuel supply passage **804**. The central passage **802** and the fuel supply passage **804** can be coaxial with one another, however this need not be the case. In the illustrated example, the fuel nozzle assembly **800** includes three supply lines, including an outer supply **806** provided by a swirler **808**, a fuel supply **810** provided by the fuel supply passage **804**, and a central supply **812** provided by the central passage **802**. In a non-limiting example, air can be provided within the central passage **802**, while other fluids or materials are contemplated, such as fuel, fuel mixes, or diluents. The fluid within the central passage **802** can have a swirl or helical component, such as imparted by a swirler or vane, or can be provided as non-swirling or laminar flow, in additional non-limiting examples.

Each of the central supply **812** and the outer supply **806** can be either swirling or non-swirling. In one non-limiting example, the outer supply **806**, the fuel supply **810**, and the central supply **812** can be imparted with a swirling component in the same direction in order to decrease shear between the different flows. Alternatively, only the central passage **802** can be non-swirling, which can provide for moving any recirculation zone aft or downstream of the end of the fuel supply passage **804**. This in turn can mitigate flame holding and flashback. In another non-limiting example, fuel provided from the fuel supply **810** can be provided between outer swirling air in the outer supply **806** and inner non-swirling air within the central passage **802**. This provides for improved fuel and air mixing downstream of the fuel supply passage **804**. It should be appreciated that the differing use of swirling or non-swirling airflows can be utilized to define a velocity profile for the fuel and air being provided, which can be utilized to mitigate the occurrence of flame holding or flashback at the fuel nozzle assembly **800** and this in turn permits the use of higher temperature fuels such as hydrogen.

In yet another non-limiting example, the central passage **802** can be used as a pilot or can be used for introducing other materials, such as diluents for NO_x suppression in one non-limiting example. If the central passage **802** is used as a pilot or diluent injection, flow through this circuit can be independently controlled for different operating cycles.

In another non-limiting example, the central passage **802** exit can be truncated to the fuel nozzle cap **708**, truncated at fuel nozzle lip **744**, can be truncated between the fuel nozzle cap **708** and the lip **744**, or any combination thereof. It is also possible that central passage **802** can be extended aft of fuel nozzle lip **744**. The passage of the central passage **802** can be made converging, constant area, or diverging to control a velocity profile at exit of the central passage **802**. The exit of the fuel nozzle lip **744** can also have features with constant area section followed by diverging section. It is contemplated that the central passage **802** can have nozzle

cap with orifice holes either axial or tangential followed by a fuel nozzle lip. Instead of orifice holes, it is possible that the central passage **802** can have a swirler, such as a swirler vane, with a low swirl number ranging from 0 to 0.5, for example.

The radial placement of a splitter **816** can control flow area and velocity at an exit of the swirler passage before air flow interacts with fuel flow. The radial location of the splitter **816** can be from 20% to 80% of the passage height "PH" that can be defined as the radial distance between a fuel nozzle **818** and a swirler outer wall **814**. In one example, the splitter **816** can be positioned from 0.2H to 5H, while other or larger ranges are contemplated. Sufficient length is provided between an aft end of the splitter **816** and a fuel nozzle aft tip **820** so that wakes generated by the splitter **816** are reduced or eliminated before the air flow interacts with the fuel flow. The aft end of the splitter **816** can have sharp vertical or radial cut, slot, or recess to minimize splitter wake, or it can include a fillet to minimize wakes from the splitter **816**.

FIG. **17** illustrates a nozzle tip **852** for a fuel nozzle **850** provided downstream from a nozzle cap **854** with the nozzle tip **852** including a diverging geometry. The nozzle tip **852** can include a lip length LL, defined along the fuel nozzle **850** from the nozzle cap **854** to the distal end of the nozzle tip **852**. A diameter D can be defined as the diameter for the fuel nozzle **850**, which increases as the nozzle tip **852** diverges at a diverging portion **860**. While the diameter D is not fully shown in the view shown in FIG. **6**, it should be appreciated that the fuel nozzle **850** can be annular or cylindrical, defining the diameter D. An opening diameter d can be defined as the diameter for openings **856** provided in the nozzle cap **854**. The ratio of the lip length LL to the diameter of the openings d can be between zero to fifty (0-50) in a non-limiting example. The ratio of the lip length LL to the nozzle diameter D can be between zero to five (0-5) in a non-limiting example. It will be understood that wider ranges are contemplated. An outer surface **858** for the nozzle tip **852** can have a constant diameter to maintain high velocity on the exterior of the fuel nozzle **850**. In one example, the fuel pressure ratio across the openings **856** can be from 1.0 to 1.4 for hydrogen fuels or hydrogen fuel blends, while other pressure ratios are contemplated based upon the particular fuel.

FIG. **18** illustrates a nozzle tip **872** for a fuel nozzle **870** provided downstream of a splitter **874** provided within an exterior swirler **876**. An outer surface **878** of the nozzle tip **872** can define a decreasing thickness **880** for the nozzle tip **872** extending aft, defining a diverging portion **882** for a radially outer swirler passage **884**. In a non-limiting example, the outer surface **878** of the nozzle tip **872** can be curved. The diverging portion **882** can begin aft of the splitter **874** in order to create a velocity profile for the air from the swirler **876** with a high axial velocity component prior to the diverging portion **882**. The diverging portion **882** can provide for decreasing or eliminating the occurrence of recirculation zones or flame holding at the nozzle tip **872**, as the inward curvature accelerates the air flow over the nozzle tip **872** to reduce or avoid flame holding on the nozzle tip **872**.

FIG. **19** illustrates a nozzle tip **892** for a fuel nozzle **890** provided downstream of a splitter **894** and a nozzle cap **896** contained within the fuel nozzle **890**. The nozzle tip **892** includes a first surface **898**, which may be an interior surface, and a second surface **900**, which may be an exterior surface. The first surface **898** and the second surface **900** can be curved such that a diverging passage **902** is defined for a

21

swirler **904**, and a converging passage **906** is defined for the fuel nozzle **890**. The fuel nozzle **890** can include a circumferential tip portion **908** with a constant cross-sectional area, while it is contemplated that this portion is removed such that the fuel nozzle **890** terminates at the end of the curvature of the interior surface **898**. Tip portion **908** can provide for improved velocity to eliminate flame holding, while eliminating the tip portion **908** can help reduce recirculation zones at the nozzle tip **892**.

FIG. **20** illustrates a fuel nozzle **930** having a nozzle cap **932** that is convex, relative to a flow of fuel through the fuel nozzle **930**. The curvature for the nozzle cap **932**, by way of non-limiting examples can be defined by a circular or elliptical profile, which can define a hemispherical or ellipsoidal shape for the nozzle cap **932**. It will be understood that other convex geometries are contemplated. Openings **934** are provided in the nozzle cap **932**, which can be similar to the openings **740** (FIG. **14**) as described herein, having an increasing swirling component nearer to the outer diameter of the fuel nozzle **930**. Furthermore, a centerline **936** of the openings **934** can be oriented toward a center of the hemisphere or ellipsoid or can be aligned axially relative to the fuel nozzle **930**, in addition to or without imparting swirl to the fuel passing through the nozzle cap **932** with a tangential orientation for the openings **934**. It should be appreciated that the openings **934** can be arranged axially, radially, tangential, or as compounded angles formed as combinations thereof.

FIG. **21** illustrates a fuel nozzle **950** having a nozzle cap **952** that is concave, relative to a flow of fuel through the fuel nozzle **950**. In non-limiting examples, the curvature of the nozzle cap **952** can be hemispherical or ellipsoidal, with it being understood that other geometries are contemplated. Openings **954** can be oriented in any suitable manner relative to the nozzle cap **952**. For example, the openings **954** can be aligned with a center of a hemisphere or ellipsoid defined by the curvature of the nozzle cap **952** or arranged axially relative to the fuel nozzle **950**. The openings **954** can provide for imparting a swirl to the flow of fuel through the fuel nozzle **950** although this need not be the case and it is contemplated that the openings **954** can impart no swirl. It should be appreciated that the openings **954** can be arranged axially, radially, tangential, or as compounded angles formed as combinations thereof.

FIGS. **22-33** illustrate plots showing a rate of swirl for an airflow provided by a swirler on the x-axis against a radial position on the y-axis. The dashed line positioned centrally represents a delineation between the fuel supply from the fuel nozzle and the air supply from the swirler, radially exterior of the fuel nozzle. More specifically, the amount or rate of swirl provided can be controlled based upon radial position, which can be defined by the swirler and the tangential openings in the fuel nozzle cap as described herein.

FIG. **22** illustrates a graph **1000** including an x-axis **1002** representing an increasing rate of tangential swirl, and a y-axis **1004** representing radial position, defined radially outward from a center of the fuel nozzle. Dashed line **1006** represents the radial transition from the fuel nozzle to the radially exterior swirler. The amount of swirl from the center of the fuel nozzle to about 50% of the radial extent of the fuel nozzle includes zero swirl, indicated at **1010**. The swirl defined in the outer half of the radial extent of the fuel nozzle, indicated at **1012**, can increase at a rate sufficient to be about the same as the rate of swirl at the radial interior of the swirler. The rate of swirl for the swirler, indicated at **1012**, can then increase at a rate higher than the swirl of the

22

fuel nozzle at **1010**, extending radially outward. Although half or fifty percent transitioning between axial to tangential direction of fuel is mentioned above by way of non-limiting example, it will be understood that this transitioning can take place at any radial location of the fuel nozzle tip.

Having no swirl within the radial center of the fuel nozzle provides for positioning a recirculation bubble in the aft direction, which can be a recirculating flow of fuel resultant of the wake generated by the fuel, which mitigates flame holding at the fuel nozzle. Increasing the rate of swirl for the fuel, from no swirl, to matching the swirl at the radial interior of the swirler provides for reducing shear between the two flows, as well as eliminating flame holding. Finally, increasing the tangential swirl in a radially outward direction within the swirler provides for mitigating flame holding on the flare cone. It should be appreciated that the numbering, size, orientation, and placement of the openings in the fuel nozzle can be varied to achieve the desired velocity distribution at the fuel nozzle exit. Furthermore, the openings can vary in size to vary the fuel momentum profile at the nozzle exit. Such velocity distribution and momentum profiles can be tailored to the particular fuel nozzle assembly or engine to reduce recirculation, mitigate flame holding, or mitigate flashback.

FIG. **23** illustrates plot **1020** with a constant increase in rate of swirl for the fuel supply, indicated at **1022**, where the rate is zero at the center. The rate of swirl at where the fuel meets the air from the swirler, indicated at **1024**, can be similar to reduce shear between the two flows.

FIG. **24** shows another plot **1030** which illustrates zero swirl for the fuel nozzle, indicated at **1032**. The rate of swirl for the swirler at the radial interior can be zero, matching that of the fuel nozzle. The rate of swirl for the swirler then has a rate increase at a constant rate extending radially outward, indicated at **1034**. No swirl for the fuel nozzle and the swirler at the radial interior provides for reduced shear between the two flows, while increasing the velocity component of the fuel. Additionally, the higher swirl on the radial exterior of the swirler prevents flame holding on the flare cone. FIG. **25** illustrates a plot **1040** where the zero-swirl extends into the swirler, such that the swirler includes no swirl on a radially interior portion, indicated at **1042**, and a swirling profile on a radially exterior portion, indicated at **1044**, which can be permitted by the splitter as described herein, separating the swirler into two flows.

FIG. **26** illustrates plot **1050** indicating, at **1052**, the center of the fuel nozzle includes zero swirl. The swirl for the radial exterior portion of the fuel nozzle can increase extending radially outward as indicated at **1054**. The swirl for the radially interior portion of the swirler, indicated at **1056**, can be the same as that of the radially outer swirl from the fuel nozzle to reduce shear between the two flows. The swirl can then decrease to zero, indicated at **1058**. The zero flow at the fuel center pushes back the recirculation bubble, while the zero swirl at the radially outer portion can increase the velocity profile along the flare cone, which can mitigate flame holding. This type of distribution is targeted for geometry with lower flare cone angle or no cylindrical flare although it is not so limited.

FIG. **27** illustrates plot **1062** having a non-zero swirl at a center of the fuel nozzle, decreasing radially outwardly to zero, indicated at **1060** and staying at zero for a portion before reaching the radial exterior of the fuel nozzle. The swirler can then include zero swirl at the radial interior, to reduce shear between the two flows, with a radially interior portion of the swirler including zero swirl, with increasing swirl extending radially outward in a radially outer portion

23

of the swirler, indicated at **1064**. The increased swirl at the radial exterior can mitigate flame holding along the flare cone.

FIG. **28** illustrates plot **1070**, which shows a non-zero swirl at a center of the fuel nozzle, decreasing radially outwardly to zero swirl at the radial exterior of the fuel nozzle, indicated at **1072**. The swirler can then include zero swirl at the radial interior, to reduce shear between the two flows, with increasing swirl extending radially outward, indicated at **1074**. The increased swirl at the radial exterior can mitigate flame holding along the flare cone.

FIG. **29** illustrates plot **1080**, which indicates a common distribution profile between the fuel nozzle and the swirler, where each includes a radially interior portion with zero swirl, indicated at **1082** and **1086**, respectively. A radially exterior portion for each of the fuel nozzle and swirler can include an increasing swirl from zero, extending radially outward, indicated at **1084** and **1088**, respectively. FIG. **30** illustrates plot **1090**, which indicates a common distribution profile between the fuel nozzle and the swirler, with each increasing from zero swirl extending radially outward, indicated at **1092** and **1094**, respectively.

FIG. **31** illustrates plot **1100** illustrating the fuel nozzle can have increasing swirl, from zero swirl at the center, increasing radially outward, indicated at **1102**. The swirl in the swirler can be constant and non-zero, which can be the same as the swirl at the radial exterior of the fuel nozzle to reduce shear between the two flows.

FIG. **32** illustrates plot **1110** where the fuel nozzle swirl increases from zero extending radially outward at a constant rate, indicated at **1112**. The swirl at the radial interior of the swirler can be the same as at of the fuel nozzle at the radial exterior, such that the shear between the two flows is reduced. The swirl in the swirler can then decrease to zero extending in the radially outward direction, indicated at **1114**. FIG. **32** illustrates another plot **1140**, where both the fuel nozzle and the swirler include a constant, non-zero swirl, indicated at **1142** and **1144**, which can reduce shear between the two flows.

FIG. **33** illustrates plot **1120** with common profiles for the fuel nozzle and the swirler. Each includes a non-zero swirler at the radial interior, or the center of the fuel nozzle, decreasing to zero swirl, indicated at **1122** and **1124**, respectively. Then, each includes a portion that increases from zero to match the swirl at the radial interior, indicated at **1126** and **1128**, respectively. In this way, shear is reduced between the fuel nozzle and the swirler, while defining variable profiles to mitigate flame holding.

While each profile in FIGS. **22-33** is shown as linear, or constant, it will be understood that the profiles could alternatively be non-constant rates of change in the swirl, such that the plots are curved. Further still, constant and non-constant combinations are contemplated. It should be further appreciated that varying the rate of swirl for the fuel flow from the fuel nozzle and the air flow from the swirler can be utilized to develop complex velocity profiles, which can be tailored to mitigate flame holding, flashback, and recirculation at various radial positions for the fuel nozzle assembly, which can provide for the use of faster-burning or higher-temperature fuels, such as hydrogen or hydrogen mixes.

Furthermore, it should be appreciated that fuel swirl or tangential angle of the holes should be gradually increased to mitigate flame on the fuel nozzle lip. The average swirl from the fuel circuit at the nozzle tip can be from 0 to 1.5, and the average swirl from the swirler air circuit exit, before it interacts with fuel, can be from 0 to 1.5. Reducing the shear between the swirler air circuit and the fuel nozzle

24

provides for a consistent velocity profile, which mitigates flame holding on the hardware downstream.

Further, it can be desirable that the fuel nozzle tangential velocity can be gradually increased to reduce a sudden deficit in velocity, pressure, or flow condition leading to high shear layer with fuel passage downstream fuel nozzle. Similarly, air flow velocity has to be well controlled to mitigate flame holding on the swirler component like splitter, swirler outer wall, swirler inner wall, and flare cone.

It should be appreciated that the aspects and embodiments provided herein are not limited to those embodiments as shown. More specifically, one or more aspects of one embodiment can be combined with, interchanged with, or removed from one or more of the other embodiments, such that additional embodiments are contemplated within the scope of this disclosure by a person having ordinary skill in the art, while not explicitly shown.

This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Further aspects of the present disclosure are provided by the subject matter of the following clauses.

A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber having a burner length L and a burner dome height H , the combustion chamber configured to combust a mixture of the hydrogen fuel flow and the compressed air flow, and the combustion chamber being characterized by a combustor size rating between one inch and seven inches.

The gas turbine engine of the preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein H is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.

25

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.

The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.

The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein L is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

The gas turbine engine of any preceding clause, further comprising one or more rotating blades.

The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.

The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.

The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.

The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.

The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plu-

26

ality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.

The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.

An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.

The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.

A gas turbine engine including a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor including a combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter.

The gas turbine engine of any preceding clause, wherein the core air flow parameter is a relationship between the thrust and bypass ratio.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

The gas turbine engine of any preceding clause, wherein the combustor size rating is based on a thrust of the gas turbine engine.

The gas turbine engine of any preceding clause, wherein the thrust is between sixty kN and five hundred kN.

The gas turbine engine of any preceding clause, wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height.

The gas turbine engine of any preceding clause, further comprising a turbine nozzle downstream of the combustion chamber, wherein the burner length is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.

The gas turbine engine of any preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein the burner dome height is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.

The gas turbine engine of any preceding clause, further comprising one or more rotating blades.

The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.

The gas turbine engine of any preceding clause, further comprising one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.

The gas turbine engine of any preceding clause, further comprising a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.

The gas turbine engine of any preceding clause, further comprising a plurality of fan blades located forward of the combustor in a puller configuration.

The gas turbine engine of any preceding clause, further comprising a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine comprising an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine comprising a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine comprising an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.

The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.

An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.

The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.

A gas turbine engine includes a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow, a compressor section configured to compress air flowing therethrough to provide a compressed air flow, and a combustor section having a fuel nozzle assembly and a combustor with a combustion characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN, wherein the combustor size rating is a function of the core air flow parameter, and wherein the combustor size rating is defined by a relationship of the burner length, squared, and the burner dome height and wherein the core air flow parameter is a relationship between the thrust and bypass ratio, and the fuel nozzle assembly including a fuel nozzle defining a longitudinal axis and a radial axis orthogonal thereto, and the fuel nozzle including a fuel passage terminating at an outlet, the fuel nozzle including nozzle tip, and a nozzle cap including a set of openings and provided within the fuel passage; wherein at least one opening of the set of openings is has a centerline oriented at a tangential angle relative to the radial axis.

The turbine engine of any preceding clause, wherein the set of openings are arranged into rows defined circumferentially relative to the longitudinal axis.

The turbine engine of any preceding clause, wherein the tangential angle for the rows of openings increases as a radial distance from the longitudinal axis increases.

The turbine engine of any preceding clause, wherein the tangential angle for an opening of the set of openings at a center of the nozzle cap is zero.

The turbine engine of any preceding clause, further including a nozzle lip defined between the nozzle cap and the nozzle tip.

The turbine engine of any preceding clause, wherein the nozzle lip includes an axial portion and a diverging portion.

The turbine engine of any preceding clause, wherein the nozzle tip defines a lip length and the fuel nozzle defines a diameter, and a ratio of lip length to diameter is between zero and five.

The turbine engine of any preceding clause, wherein the nozzle tip defines a lip length and each opening of the set of openings defines an opening diameter, and a ratio of lip length to opening diameter is between zero and fifty.

The turbine engine of any preceding clause, wherein the nozzle lip includes a diverging portion defined on an exterior surface of the fuel nozzle.

The turbine engine of any preceding clause, wherein the nozzle lip further includes a diverging portion defined on an interior surface of the fuel nozzle.

The turbine engine of any preceding clause, wherein the nozzle cap is curved as concave or convex relative to a flow direction through the fuel passage.

The turbine engine of any preceding clause, further including a central passage extending within the fuel passage and extending through the nozzle cap.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

The gas turbine engine of any preceding clause, wherein the combustor size rating is based on a thrust of the gas turbine engine.

The gas turbine engine of any preceding clause, wherein the thrust is between sixty kN and five hundred kN.

The gas turbine engine of any preceding clause, further including a turbine nozzle downstream of the combustion chamber, wherein the burner length is the distance between a plane orthogonal to a forward line at which the burner dome height is measured and a leading edge of the turbine nozzle.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and thirty-five square inches.

The gas turbine engine of any preceding clause, wherein the combustor further includes an outer liner and an inner liner, the combustion chamber having a forward end and being defined between the outer liner and the inner liner, each of the outer liner and the inner liner having an inner surface, and wherein the burner dome height is the maximum height between the inner surface of the outer liner and the inner surface of the inner liner at the forward end of the combustion chamber.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and seven inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the combustor size rating is between three inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two inches and three inches.

The gas turbine engine of any preceding clause, wherein the burner length is between two and one half inches and three and one half inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and six inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between two and one half inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner dome height is between four inches and five inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twenty square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between six square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein the burner length, squared, is between eight square inches and twelve square inches.

The gas turbine engine of any preceding clause, wherein no diluent is added to the combustion chamber.

The gas turbine engine of any preceding clause, further including one or more rotating blades.

The gas turbine engine of any preceding clause, further including one or more stationary vanes, wherein the one or more stationary vanes are positioned forward of the rotating blades.

The gas turbine engine of any preceding clause, further including one or more stationary vanes, wherein the one or more stationary vanes are positioned aft of the rotating blades.

The gas turbine engine of any preceding clause, further including a first set of one or more rotating blades and a second set of rotating blades, the first set of rotating blades and the second set of rotating blades being configured to operate in a counter-rotating fashion.

The gas turbine engine of any preceding clause, further including a plurality of fan blades located forward of the combustor in a puller configuration.

The gas turbine engine of any preceding clause, further including a plurality of fan blades, the combustor located forward of the plurality of fan blades in a pusher configuration.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is one of a turbofan engine, an unducted single fan engine, a turbojet engine, a turboshaft engine, or a turboprop engine.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbofan engine including an outer nacelle that houses the compressor section, the combustor, and a plurality of fan blades.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is an unducted single fan engine including a spinner coupled to a nacelle, the nacelle housing the compressor section and the combustor, a plurality of outlet guide vanes coupled to an outer surface of the nacelle, and a plurality of fan blades coupled to the spinner and rotatable therewith.

The gas turbine engine of any preceding clause, wherein the gas turbine engine is a turbojet engine including an outer nacelle that houses the compressor section and the combustor, the turbojet engine not including a fan with bypass duct.

The gas turbine engine of any preceding clause, further including a hydrogen fuel tank for holding the hydrogen fuel in a liquid phase, the hydrogen fuel delivery assembly being connected to the hydrogen fuel tank, and a vaporizer in communication with the hydrogen fuel delivery assembly for heating the hydrogen fuel in the liquid phase to at least one of a gaseous phase and a supercritical phase, the vaporizer being located between the hydrogen fuel tank and the combustor.

An aircraft including a fuselage, a wing connected to the fuselage, and the gas turbine engine of any preceding clause.

31

The aircraft of the preceding clause, wherein the hydrogen fuel tank is positioned at least partially within at least one of the fuselage and the wing, and wherein the vaporizer is positioned at least partially within at least one of the fuselage, the wing, and the gas turbine engine.

A method of providing fuel and air to a turbine engine of any proceeding clause, the method including providing a supply of fuel via a fuel nozzle defining a longitudinal axis, and a supply of air via a swirler circumscribing the fuel nozzle and imparting a tangential component, tangent to a radius extending from the longitudinal axis, to the supply of fuel with a set of openings provided in a nozzle cap arranged at a tangential angle.

The method of any preceding clause, wherein the tangential component imparted to the supply of fuel is complementary to a swirl imparted by the swirler to the supply of air.

The method of any preceding clause, wherein the swirler includes a splitter, separating the supply of air into a radially inner supply and a radially outer supply, and the tangential component imparted to the supply of fuel is complementary to the swirl imparted to the radially inner supply.

The method of any preceding clause, wherein the tangential component imparted to the supply of fuel is counter rotating relative to the supply of air provided by the swirler.

Although the foregoing description is directed to the preferred embodiments, it is noted that other variations and modifications will be apparent to those skilled in the art and may be made without departing from the spirit or scope of the disclosure. Moreover, features described in connection with one embodiment may be used in conjunction with other embodiments, even if not explicitly stated above.

What is claimed is:

1. A gas turbine engine comprising:

a hydrogen fuel delivery assembly configured to deliver a hydrogen fuel flow;

a compressor section configured to compress air flowing therethrough to provide a compressed air flow; and

a combustor section including a fuel nozzle assembly and a combustor configured to operate without diluent and including an inner liner, outer liner, and a combustion chamber, and the combustion chamber characterized by a combustor size rating between one inch and seven inches at a core air flow parameter between two and one half kN and sixty kN,

wherein the combustor size rating is a function of the core air flow parameter, and

wherein the combustor size rating is defined by:

$$\frac{L^2}{H}$$

wherein H is a maximum height of the combustion chamber measured by a forward line extending from an inner surface of the outer liner to an inner surface of the inner liner and L is a length measured from a midpoint of the forward line to a midpoint of an aft line, the aft line extending from the inner surface of the inner liner to the inner surface of the outer liner at a leading edge of a turbine nozzle,

and, wherein the core air flow parameter is defined by:

$$\frac{\text{Thrust}}{\text{Bypass Ratio}}$$

32

and, the fuel nozzle assembly comprising:

a fuel nozzle defining a longitudinal axis and a radial axis orthogonal thereto, and the fuel nozzle including a fuel passage terminating at an outlet, the fuel nozzle including a nozzle tip; and

a nozzle cap including a set of openings and provided within the fuel passage, wherein at least one opening of the set of openings has a centerline oriented at a tangential angle relative to the radial axis.

2. The gas turbine engine of claim 1, wherein the combustor size rating is between two inches and three and one quarter inches at a core air flow parameter between two and one half kN and fifty kN.

3. The gas turbine engine of claim 1, wherein the combustor size rating is based on a thrust of the gas turbine engine.

4. The gas turbine engine of claim 3, wherein the thrust is between sixty kN and five hundred kN.

5. The gas turbine engine of claim 1, wherein the set of openings are arranged into rows defined circumferentially relative to the longitudinal axis.

6. The gas turbine engine of claim 1, wherein the at least one opening includes another opening at a center of the nozzle cap, wherein the centerline of the another opening is oriented at zero degrees relative to the longitudinal axis.

7. The gas turbine engine of claim 1, further comprising a nozzle lip defined between the nozzle cap and the nozzle tip.

8. The gas turbine engine of claim 7, wherein the nozzle lip includes an axial portion and a diverging portion.

9. The gas turbine engine of claim 7, wherein the nozzle tip defines a lip length and the fuel nozzle defines a nozzle diameter, and a ratio of lip length to nozzle diameter is less than five.

10. The gas turbine engine of claim 7, wherein the nozzle tip defines a lip length and each opening of the set of openings defines an opening diameter, and a ratio of lip length to opening diameter is between less than or equal to fifty.

11. The gas turbine engine of claim 7, wherein the nozzle lip includes a first diverging portion defined on an exterior surface of the fuel nozzle, a second diverging portion defined on an interior surface of the fuel nozzle, or a combination thereof.

12. The gas turbine engine of claim 1, wherein the nozzle cap is curved as concave or convex relative to a flow direction through the fuel passage.

13. The gas turbine engine of claim 1, further comprising a central passage extending within the fuel passage and extending through the nozzle cap.

14. The gas turbine engine of claim 5, wherein the set of openings include the at least one opening and multiple openings, wherein the multiple openings include a respective centerline and a respective radius axis perpendicular to the longitudinal axis, wherein each of the multiple openings of the set of openings have a respective tangential angle measured between the respective radius axis and the respective centerline.

15. The gas turbine engine of claim 14, wherein the respective tangential angle of the multiple openings of the set of openings increases as a radial distance from the longitudinal axis increases.